eLife

Immunology and Inflammation

RBP-J regulates homeostasis and function of circulating Ly6C^{lo} monocytes

Reviewed Preprint

Revised by authors after peer review.

About eLife's process

Reviewed preprint version 2
February 13, 2024 (this version)

Reviewed preprint version 1
June 7, 2023

Posted to preprint server
April 14, 2023

Sent for peer review
April 13, 2023

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Abstract

Notch-RBP-J signaling plays an essential role in maintenance of myeloid homeostasis. However, its role in monocyte cell fate decisions is not fully understood. Here we showed that conditional deletion of transcription factor RBP-J in myeloid cells resulted in marked accumulation of blood Ly6C^{lo} monocytes that highly expressed chemokine receptor CCR2. Bone marrow transplantation and parabiosis experiments revealed a cell intrinsic requirement of RBP-J for controlling blood Ly6C^{lo}CCR2^{hi} monocytes. RBP-J-deficient Ly6C^{lo} monocytes exhibited enhanced capacity competing with wildtype counterparts in blood circulation. In accordance with alterations of circulating monocytes, RBP-J deficiency led to markedly increased population of lung tissues with Ly6C^{lo} monocytes and CD16.2^{+} interstitial macrophages. Furthermore, RBP-J deficiency-associated phenotypes could be genetically corrected by further deleting *Ccr2* in myeloid cells. These results demonstrate that RBP-J functions as a crucial regulator of blood Ly6C^{lo} monocytes and thus derived lung-resident myeloid populations, at least in part through regulation of CCR2.

eLife assessment

This study presents a **valuable** examination into the role Notch-RBP-J signalling in regulating monocyte subset homeostasis. The data were collected and analysed using **solid** and validated methodology and can be used as a starting point for exploring the mechanisms involved in RBP-J signalling in non-classical monocytes. The data presented strongly confirm the authors conclusions. However, this paper primarily focuses on providing a description, and additional studies are necessary to fully elucidate the mechanisms through which RBP-J deficiency contributes to the specific increase in Ly6Clo monocyte numbers in both the blood and lungs.

Introduction

Monocytes are integral components of the mononuclear phagocyte system (MPS) that develop from monocyte precursors in the bone marrow(Guilliams et al., 2018). Several functionally and phenotypically distinct subsets of blood monocytes have been defined on the basis of expression of surface markers(Cros et al., 2010; Geissmann et al., 2003; Passlick et al., 1989; Weber et al., 2000). In mice, Ly6C^{hi} monocytes (also called inflammatory or classical monocytes) characterized by Ly6C^{hi}CCR2^{hi}CX3CR1^{lo} exhibit a short half-life, and are recruited to tissue and differentiate into macrophages and dendritic cells(Ginhoux and Jung, 2014). Ly6C^{hi} monocytes are analogous to CD14^{+} human monocytes based on gene expression profiling(Ingersoll et al., 2010). A second subtype of monocytes is called Ly6C^{lo} monocytes (also termed patrolling monocytes or non-classical monocytes), which express high level of CX3CR1 and low level of CCR2, equivalent to CD14^{lo}CD16^{+} human monocytes. Ly6C^{lo} monocytes are regarded as blood-resident macrophages that patrol blood vessels and scavenge microparticles attached to the endothelium under physiological conditions, and exhibit a long half-life in the steady-state(Auffray et al., 2007; Carlin et al., 2013; Yona et al., 2013). Ly6C^{lo} monocytes may exert a protective effect by suppressing atherosclerosis in mice, while high levels of non-classical monocytes have been associated with more advanced vascular dysfunction and oxidative stress in patients with coronary artery disease(Hamers et al., 2012; Hanna et al., 2012; Urbanski et al., 2017). In the inflammatory settings, Ly6C^{lo} monocytes often show anti-inflammatory properties, yet also exhibit a proinflammatory role under certain circumstances(Amano et al., 2005; Brunet et al., 2016; Carlin et al., 2013; Cros et al., 2010; Misharin et al., 2014; Nahrendorf et al., 2007; Puchner et al., 2018). In addition, Ly6C^{lo} monocytes have demonstrated protective functions in tumorigenesis, such as engulfing tumor materials to prevent cancer metastasis, activating and recruiting NK cells to the lungs(Hanna et al., 2015; Thomas et al., 2016). While the functions of Ly6C^{lo} monocytes are complex and sometimes contradictory depending on the animal models and experimental approaches, it is clear that they play a crucial role in both health and disease.

The generation and maintenance of Ly6C^{lo} monocytes are regulated by several transcription factors, including Nr4a1 and CEBPβ(Hanna et al., 2011; Tamura et al., 2017). The colony-stimulating factor 1 receptor (CSF1R) signaling pathway is important for the generation and survival of Ly6C^{lo} monocytes, and neutralizing antibodies against CSF1R can reduce their numbers(MacDonald et al., 2010). Interestingly, recent studies have suggested that different subsets of monocytes may be supported by distinct cellular sources of CSF-1 within bone marrow niches, in which targeted deletion of *Csf1* from sinusoidal endothelial cells selectively reduces Ly6C^{lo} monocytes but not Ly6C^{hi} monocytes(Emoto et al., 2022). LAIR1 has been shown to be activated by stromal protein Colec12, and LAIR1 deficiency leads to aberrant proliferation and apoptosis in bone marrow non-classical monocytes(Keerthivasan et al., 2021). These findings further underscore the complexity of the mechanisms that regulate Ly6C^{lo} monocytes, and suggest that multiple factors are involved in this process.

Recombination signal-binding protein for immunoglobulin kappa J region (RBP-J; also named CSL in humans) is commonly known as the master nuclear mediator of canonical Notch signaling(Radtke et al., 2013). In the absence of Notch intracellular domain, RBP-J associates with co-repressor proteins to repress transcription of downstream target genes. In the immune system, the best studied functions of Notch signaling are its roles in regulating the development of lymphocytes, such as T cells and marginal zone B cells(Radtke et al., 2013). Notch signaling also has been reported to regulate the differentiation and function of myeloid cells including granulocyte/monocyte progenitors, osteoclasts and dendritic cells (Caton et al., 2007; Klinakis et al., 2011; Lewis et al., 2011; Zhao et al., 2012). In osteoclasts, RBP-J represses osteoclastogenesis, particularly under inflammatory conditions(Zhao et al., 2012). In dendritic cells, Notch-RBP-J signaling controls the maintenance of CD8^{-} DC(Caton et al., 2007). In addition, Notch2-RBP-J pathway is essential for development of CD8^{-}ESAM^{+} DC in the spleen and CD103^{+}CD11b^{+} DC in the lamina propria of the intestine(Lewis et al., 2011). However, the role of

Notch-RBP-J signaling pathway in regulating monocyte subsets remains elusive. Here, we demonstrated that RBP-J controlled homeostasis of blood Ly6C^{lo} monocytes in a cell intrinsic manner. RBP-J deficiency led to a drastic increase in blood Ly6C^{lo} monocytes, lung Ly6C^{lo} monocytes and CD16.2^{+} IM. Our results suggest that RBP-J plays a critical role in regulating the population and characteristics of Ly6C^{lo} monocytes in the blood and lung.

Results

RBP-J is essential for maintenance of blood Ly6Clo monocytes

To investigate the role of RBP-J in monocyte subsets, we utilized mice with RBP-J specific deletion in the myeloid cells (*Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$ mice). Efficient deletion of *Rbpj* in blood monocytes was confirmed by quantitative real-time PCR (qPCR) (**Figure 1** – figure supplement 1A). Flow cytometric analysis revealed that RBP-J deficient mice had a significant increase in the proportion of Ly6C^{lo} monocytes, but not Ly6C^{hi} monocytes, in blood, compared to age-matched control mice with the genotype *Rbpj*$^{+/+}$*Lyz2*$^{cre/cre}$ (**Figure 1A**). In contrast to circulating monocytes, RBP-J deficient mice exhibited minimal alterations in the percentages of monocyte subsets in bone marrow (BM) and spleen (**Figure 1B** and **Figure 1** – figure supplement 1C). Additionally, RBP-J deficiency in myeloid cells did not appear to have an effect on neutrophils (Figure 1– figure supplement 1B). Therefore, among circulating myeloid populations, RBP-J selectively controlled the subset of Ly6C^{lo} monocytes.

BM progenitors that give rise to circulating monocytes are monocyte-dendritic cell progenitors (MDPs) and common monocyte progenitors (cMoPs)(Hanna et al., 2011; Hettinger et al., 2013; Liu et al., 2019; Varol et al., 2007). Next, we analyzed BM progenitors, and found that the percentages of MDPs and cMoPs were equivalent in control and RBP-J deficient mice (**Figure 1C**). These results in conjunction with the observations of normal BM monocyte populations suggest that alterations of peripheral blood Ly6C^{lo} monocytes may not originate from BM. Next, we examined whether RBP– J may influence the egress of Ly6C^{lo} monocytes from BM, and measured BM exit rate of Ly6C^{lo} monocytes using *in vivo* labeling of sinusoidal cells as previously described (**Figure 1D**)(Debien et al., 2013). The results showed that RBP-J deficiency did not affect egress of Ly6C^{lo} monocytes from BM (**Figure 1E**). Taken together, these data revealed RBP-J as a critical regulator controlling homeostasis of peripheral blood Ly6C^{lo} monocytes.

RBP-J deficiency does not affect Ly6C^{hi} monocyte conversion or Ly6C^{lo} monocyte survival and proliferation

Next, we aimed to determine whether the increase of blood Ly6C^{lo} monocytes in RBP– J deficient mice was due to decreased cell death or enhanced proliferation. We first stained monocytes with Annexin V and 7-AAD to identify apoptotic cells and observed comparable percentages of apoptotic blood Ly6C^{lo} monocytes in control and RBP-J deficient mice (**Figure 2A**). Given the crucial role of Nr4a1 in the survival of Ly6C^{lo} monocytes(Hanna et al., 2011), we detected the expression of Nr4a1, which was similar in Ly6C^{lo} monocytes from control and RBP-J deficient mice (Figure 2 – figure supplement 1A). We then assessed the proliferative capacity by analyzing Ki-67 expression as well as *in vivo* EdU incorporation. Ki-67 levels in blood monocytes displayed no differences between control and RBP-J deficient mice (**Figure 2B**). The percentage of EdU^{+} monocytes did not significantly differ between control and RBP-J deficient mice at the indicated time points, implying that the turnover of Ly6C^{lo} monocytes was normal in RBP-J deficient mice (**Figure 2C**). Fluorescent latex beads as particulate tracers could be phagocytosed by monocytes after intravenous injection, and stably label Ly6C^{lo} monocytes(Tacke et al., 2006). Thus, we intravenously injected fluorescent latex beads into control and RBP-J deficient mice to track circulating monocytes (**Figure 2D**). By day 7 post injection, only Ly6C^{lo} monocytes were latex^{+}

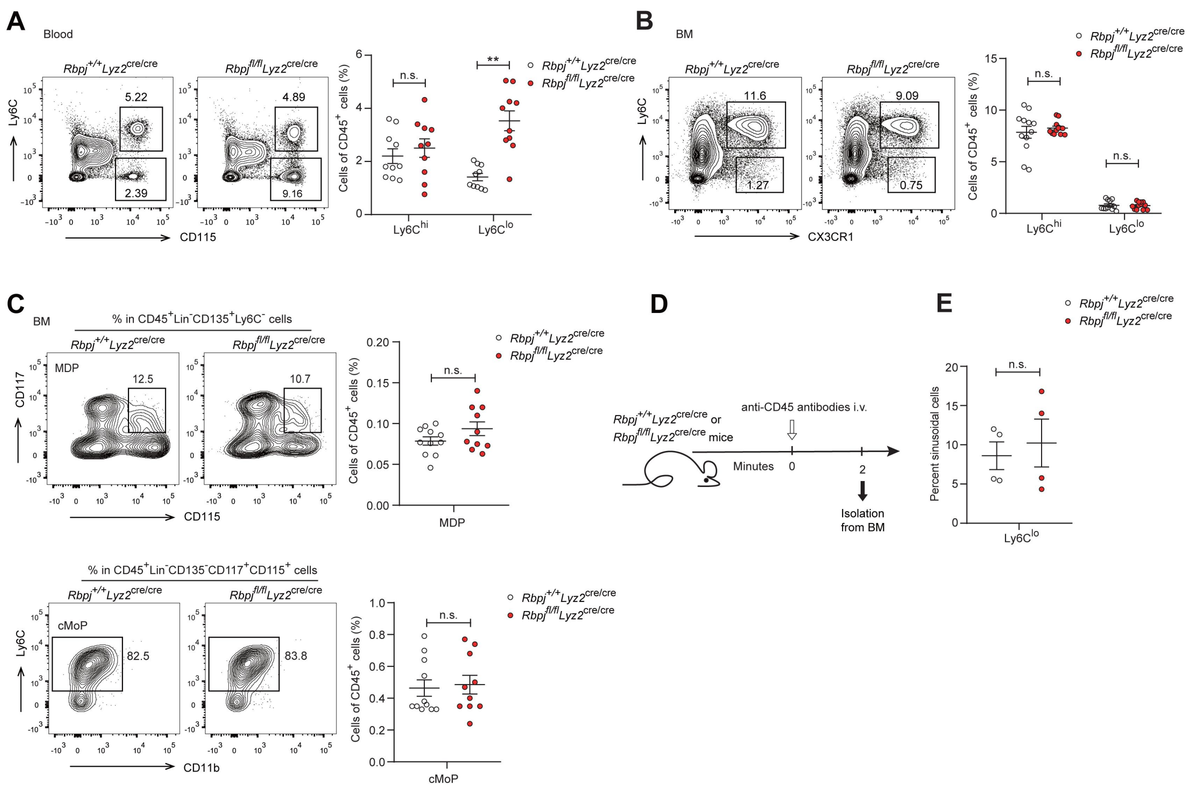

Figure 1.

RBP-J deficient mice display more Ly6C^{lo} blood monocytes.

(A) Blood Ly6C^{hi} and Ly6C^{lo} monocytes in *Rbpj$^{+/+}$Lyz2*$^{cre/cre}$ control and *Rbpj$^{fl/fl}$Lyz2*$^{cre/cre}$ mice were determined by flow cytometry analyses (FACS). Representative FACS plots (left) and cumulative data of cell ratio (right) are shown.
(B and C) Representative FACS plots and cumulative data quantitating percentages of BM monocyte subsets (B) and myeloid progenitor cells (C) (CD45^{+}CD11b^{+}Ly6G^{-} CD115^{+} Ly6C^{hi} monocyte; CD45^{+}CD11b^{+}Ly6G^{-}CD115^{+}Ly6C^{lo} monocyte; MDP, CD45^{+}Lin^{-}CD117^{+}CD115^{+}CD135^{+} Ly6C^{-}; cMoP, CD45^{+}Lin^{-}CD11b^{-}CD117^{+}CD115^{+} CD135^{-}Ly6C^{+}). Lin: CD3, B220, Ter119, and Gr-1, CD11b.
(D) Experimental outline for panel (E).
(E) Cumulative data quantitating percentages of sinusoidal monocytes (CD45^{+}) within total BM Ly6C^{lo} monocytes.
Data are pooled from at least 2 independent experiments; n ≥ 4 in each group. Data are shown as mean ± SEM; n.s., not significant; **, P < 0.01 (two-tailed Student's unpaired t test). Each symbol represents an individual mouse.

as previously reported(Tacke et al., 2006), whereas control and RBP-J deficient mice groups presented a similar frequency of latex^{+} monocytes (**Figure 2E**). Together, these data indicated that RBP-J did not influence monocyte survival and proliferation.

Previous studies have shown that Ly6C^{lo} monocytes are observed in recipient mice following the adoptive transfer of Ly6C^{hi} monocytes, indicating that Ly6C^{hi} monocytes can convert into Ly6C^{lo} monocytes(Varol et al., 2007; Yona et al., 2013). We next wished to evaluate whether conversion of Ly6C^{hi} monocyte was regulated by RBP-J by isolating bone marrow GFP^{+}Ly6C^{hi} monocytes from *Lyz2*$^{cre/cre}$*Cx3cr1*$^{gfp/+}$ control or *Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$*Cx3cr1*$^{gfp/+}$ mice and adoptively transferring them into *Ccr2*$^{RFP/RFP}$ recipients (**Figure 2F**). Sixty hours after transfer, a subset of cells from donors were converted into Ly6C^{lo} monocytes (**Figure 2** – figure supplement 1B), and equal percentages of Ly6C^{lo} monocytes were derived from control and *Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$*Cx3cr1*$^{gfp/+}$donors (**Figure 2G**). Thus, the conversion of Ly6C^{hi} monocyte into Ly6C^{lo} monocyte was not affected by RBP-J deficiency.

RBP-J regulates blood Ly6C^{lo} monocytes in a cell intrinsic manner

We next wondered whether the increase of Ly6C^{lo} monocytes in RBP-J deficient mice was bone marrow-derived and cell intrinsic. We performed bone marrow transplantation by engrafting lethally irradiated mice with a 1:4 mixture of *Rbpj*$^{+/+}$*Lyz2*$^{cre/cre}$ control and *Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$ BM cells (CD45.2) and *Cx3cr1*$^{gfp/+}$ BM cells (CD45.1) (**Figure 3A**). Eight weeks after transplantation, we analyzed the frequencies of CD45.2^{+} donor cells in the bone marrow and blood of recipient mice and found that the frequencies of CD45.2^{+} cells within total cells were similar to the mixture ratio of bone marrow cells for Ly6C^{hi} monocytes and neutrophils in both BM and blood (**Figure 3B** and **Figure 3** – figure supplement 1A and B). Specifically, more blood Ly6C^{lo}monocytes were derived from RBP-J deficient donors than control donors (**Figure 3B**), reflecting that the contribution of RBP-J deficient cells in blood Ly6C^{lo} cells was significantly higher than that of control cells. These results implied that RBP-J deficient BM cells were highly efficient in generating blood Ly6C^{lo} monocytes.

Given that RBP-J deficient mice had more Ly6C^{lo} monocytes in blood than did control mice, we performed parabiosis experiments, a surgical union of two organisms, which allowed parabiotic mice to share their blood circulation. The contribution of circulating cells from one animal in another can be estimated by measuring the percentage of blood cells that originated from each animal(Liu et al., 2007). We joined a CD45.1, *Cx3cr1*$^{gfp/+}$ mouse with an age and sex-matched *Rbpj*$^{+/+}$*Lyz2*$^{cre/cre}$ control or *Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$ mouse (CD45.2) (**Figure 3C**). Four weeks after the procedure, about 50% of B cells and T cells in parabiotic mice displayed efficient exchange of their circulation (**Figure 3** – figure supplement 1C). As expected, RBP-J deficient mice exhibited higher percentages of Ly6C^{lo} monocytes than control animals (**Figure 3D**). Intriguingly, in the parabiotic *Cx3cr1*$^{gfp/+}$mice, RBP-J deficient cells still constituted significantly higher proportion of circulating Ly6C^{lo}, but not Ly6C^{hi} monocytes, than RBP-J sufficient counterparts as indicated by the percentages of the GFP-negative population (**Figure 3E**). These results implicated that the enhanced ability of Ly6C^{lo} monocytes to circulate in the peripheral blood as a result of RBP-J deficiency was cell intrinsic.

RBP-J regulates phenotypical marker genes in blood Ly6C^{lo} monocytes

To further study the consequences of RBP-J loss of function in blood Ly6C^{lo} monocytes, we performed gene expression profiling by RNA-seq, which revealed that Ly6C^{lo} monocytes in RBP-J deficient mice exhibited low expression of *Itgax* but high expression of *Ccr2*, in comparison to those in control mice (**Figure 4A**). These differential expression patterns of phenotypic marker genes were also confirmed by qPCR and flow cytometry analysis (**Figure 4B**). However, Ly6C^{hi} blood monocytes and BM monocytes displayed normal levels of CD11c and CCR2 (**Figure 4B** and

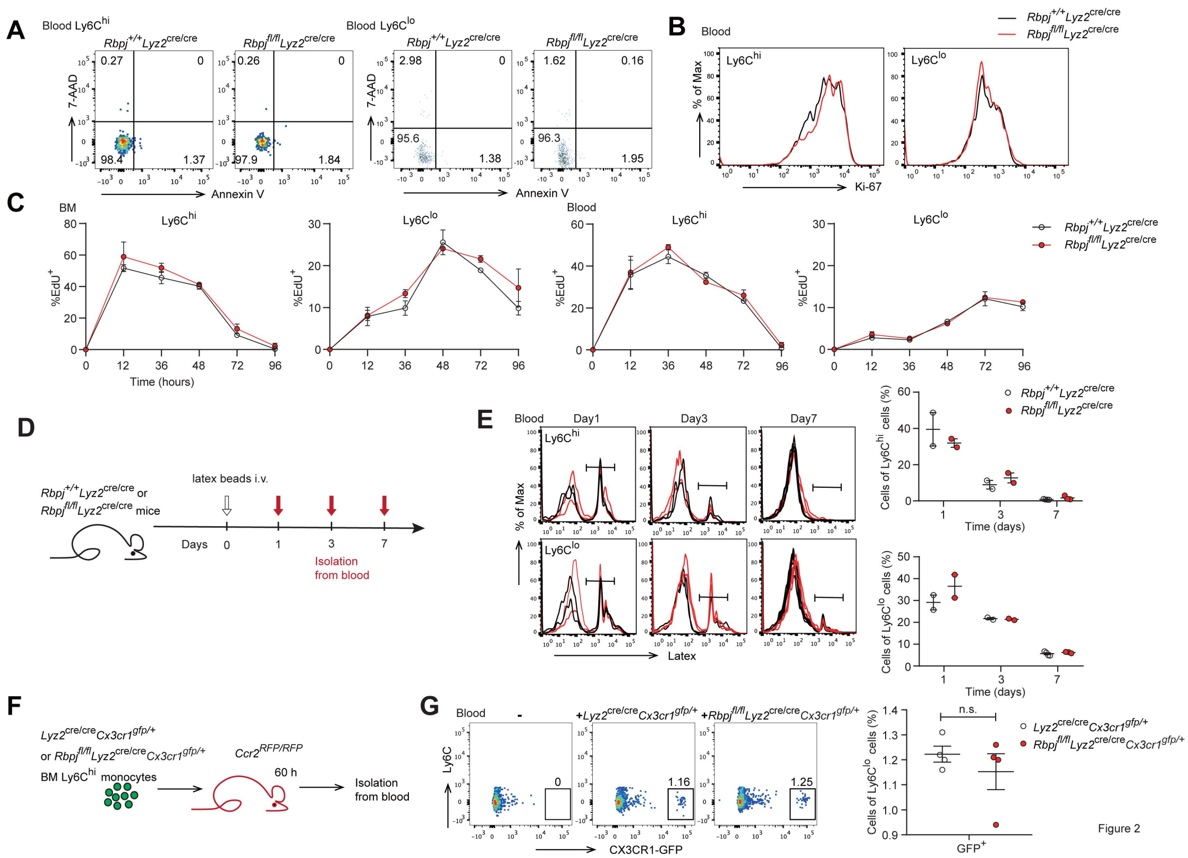

Figure 2.

Monocyte subsets in RBP-J deficient mice display normal cell death.

(A) Representative FACS plots of monocyte subsets in blood stained with 7-AAD and Annexin V.
(B) FACS analysis of Ki-67 expression in *Rbpj*$^{+/+}$*Lyz2*$^{cre/cre}$ control and *Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$ blood monocyte subsets. Black lines represent control mice and red lines represent RBP-J deficient mice.
(C) Analysis of time course of EdU incorporation of monocyte subsets in BM and blood after a single 1 mg EdU pulsing. The percentages of EdU^{+} cells among the indicated monocyte subsets are shown.
(D) Experimental outline for panel (E).
(E) Analysis of time course of latex beads incorporation of monocyte subsets in blood after latex beads injection. The percentages of latex^{+} cells among the indicated monocyte subsets are shown.
(F) Cartoon depicting the adoptive transfer. BM GFP^{+}Ly6C^{hi} monocytes were sorted from *Lyz2*$^{cre/cre}$*Cx3cr1*$^{gfp/+}$or *Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$*Cx3cr1*$^{gfp/+}$mice and transferred into *Ccr2*$^{RFP/RFP}$ recipient mice. Sixty hours after transfer, cell fate was analyzed.
(G) Representative FACS plots are shown in the left panel, and the frequencies of GFP^{+}Ly6C^{lo} monocytes within total Ly6C^{lo} monocytes are shown in the right panel.
Data are pooled from 2 independent experiments (G); n ≥ 2 in each group (C, E and G). Data are shown as mean ± SEM; n.s., not significant; (two-tailed Student's unpaired t test). Each symbol represents an individual mouse (E and G).

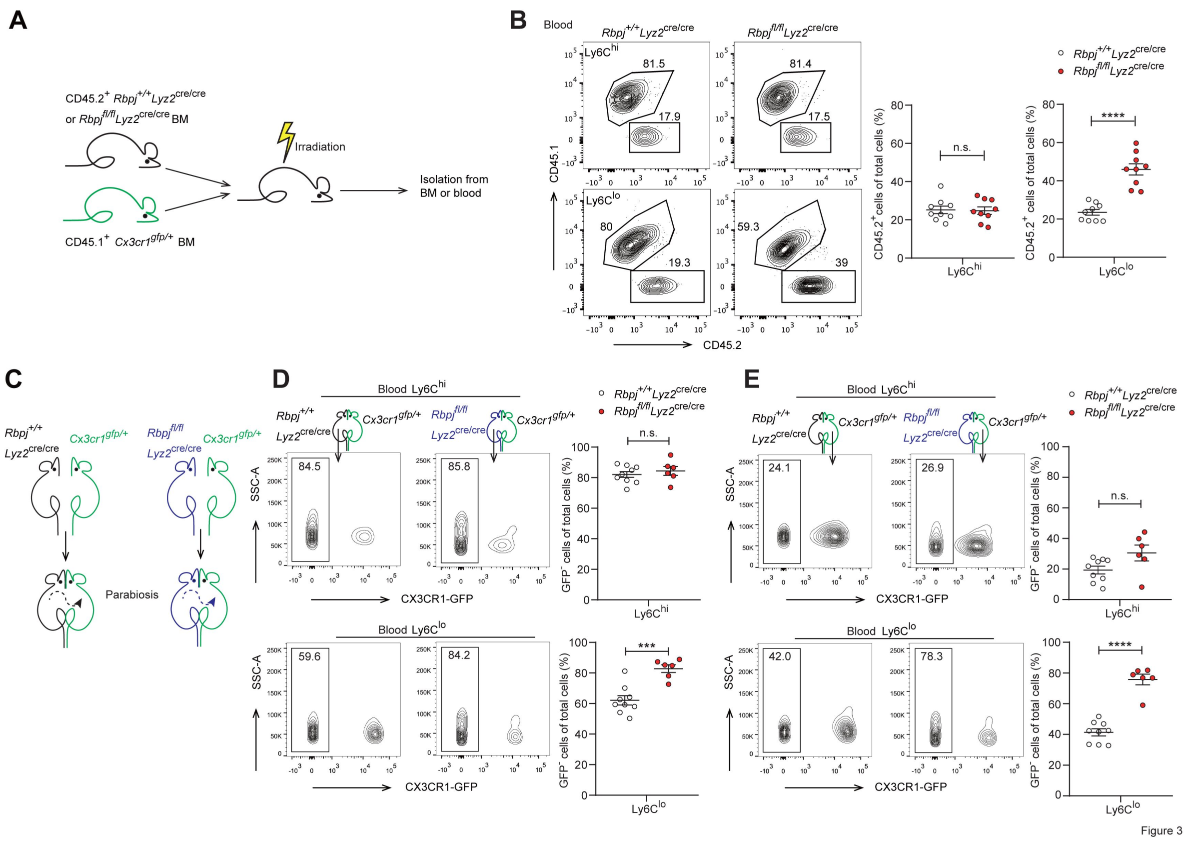

Figure 3.

The role of RBP-J in blood Ly6C^{lo} monocytes is cell intrinsic.

(A) Cartoon depicting the bone marrow transplantation. BM recipient 6-week-old male C57BL/6 mice (CD45.2) were lethally irradiated. BM cells from donor mice (*Rbpj*$^{+/+}$*Lyz2*$^{cre/cre}$ or *Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$ and CD45.1, *Cx3cr1*$^{gfp/+}$) were collected and transferred into recipient mice. Mice were used after 8 weeks of bone marrow reconstitution.
(B) Representative FACS plots (left) and cumulative data (right) quantitating the frequency of *Rbpj*$^{+/+}$*Lyz2*$^{cre/cre}$ and *Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$ donor cells among Ly6C^{hi} and Ly6C^{lo} monocytes in the blood of recipient mice.
(C) Cartoon depicting the generation of *Rbpj*$^{+/+}$*Lyz2*$^{cre/cre}$ control or *Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$ and *Cx3cr1*$^{gfp/+}$ parabiotic pairs.
(D) Representative FACS plots (left) and cumulative data (right) quantitating percentages of monocyte subsets derived from control or RBP-J deficient mice in control or RBP-J deficient mice.
(E) Representative FACS plots (left) and cumulative data (right) quantitating percentages of monocyte subsets derived from control or RBP-J deficient mice in *Cx3cr1*$^{gfp/+}$ mice.
Data are pooled from at least 2 independent experiments; n ≥ 6 in each group. Data are shown as mean ± SEM; n.s., not significant; ***, P < 0.001; ****, P < 0.0001 (two– tailed Student's unpaired t test). Each symbol represents an individual mouse. SSC-A, side scatter area.

Figure 4 – figure supplement 1A). Moreover, principal component analysis showed distinct expression pattern of monocytes between control and RBP-J deficient mice (**Figure 4C**). To determine whether the regulation of phenotypic markers by RBP-J in blood Ly6C^{lo} monocytes was cell-intrinsic, we performed BM transplantation as described above (**Figure 3A**). The findings indicated that Ly6C^{lo} monocytes derived from RBP-J deficient BM cells expressed high levels of CCR2 but low levels of CD11c, in comparison to those derived from control BM cells, whereas no significant changes were observed in blood Ly6C^{hi} and BM monocytes, as expected (**Figure 4D** and **Figure 4** – figure supplement 1B). These results collectively suggested that RBP-J regulated the expression of CCR2/CD11c in a cell-intrinsic manner.

RBP-J-mediated control of blood Ly6C^{lo} monocytes is CCR2 dependent

Given that RBP-J deficiency led to enhanced CCR2 expression in Ly6C^{lo} monocytes and that CCR2 is essential for monocyte functionality under various inflammatory and non-inflammatory conditions(Shi and Pamer, 2011), we wished to genetically investigate the role of CCR2 in the RBP-J deficient background. We generated RBP– J/CCR2 double deficient (DKO) mice with the genotype $Rbpj^{fl/fl}Lyz2^{cre/cre}Ccr2^{RFP/RFP}$ with $Lyz2^{cre/cre}Ccr2^{RFP/+}$ and $Rbpj^{fl/fl}Lyz2^{cre/cre}Ccr2^{RFP/+}$ mice serving as controls. As expected, the expression of CCR2 in Ly6C^{lo} monocytes was reduced in DKO mice compared to RBP-J deficient mice (**Figure 5A**), confirming the successful deletion of the *Ccr2* gene. DKO mice showed a lower percentage of both Ly6C^{hi} and Ly6C^{lo} monocytes than RBP-J deficient mice (**Figure 5B and C**). Notably, the percentage of Ly6C^{lo} monocytes in DKO mice was comparable to that observed in the control mice (**Figure 5B and C**), implicating that deletion of CCR2 corrected RBP-J deficiency– associated phenotype of increased Ly6C^{lo} monocytes. Whereas, both RBP-J deficient and DKO mice exhibited lower expression levels of CD11c in their Ly6C^{lo} monocyte than control mice, suggesting that the regulation of CD11c expression was independent of CCR2 (**Figure 5D**). These results suggested that RBP-J regulated Ly6C^{lo} monocytes, at least in part, through CCR2.

Lung Ly6C^{lo} monocytes are accumulated in RBP-J deficient mice

In mice, alveolar macrophages (AM) are maintained by local self-renewal, and arise from fetal liver derived precursors, whereas interstitial macrophages (IM) are probably originated from monocytes (Guilliams et al., 2013; Hashimoto et al., 2013; Sabatel et al., 2017). Schyns et al identified 3 subpopulations of IM according to expression of CD206 and CD16.2 and Ly6C^{lo} monocytes are proposed to give rise to CD64^{+}CD16.2^{+} IM(Schyns et al., 2019). We next compared the populations of lung monocytes and IM between control and RBP-J deficient mice. The conditional deletion of RBP-J resulted in a significant increase in the absolute and relative numbers of Ly6C^{lo} monocytes and CD16.2^{+} IM, while the numbers of Ly6C^{hi} monocytes, CD206^{-} IM, and CD206^{+} IM remained unchanged (**Figure 6A and B**). To confirm these findings, lung sections from $Lyz2^{cre/cre}Cx3cr1^{gfp/+}$ control and $Rbpj^{fl/fl}Lyz2^{cre/cre}Cx3cr1^{gfp/+}$ mice were stained with an anti-GFP antibody, which revealed a significant increase in the number of GFP^{+} cells in RBP-J deficient mice (**Figure 6C**). In addition, we evaluated the proliferative capacity of Ly6C^{lo} monocytes and CD16.2^{+} IM, but did not observe enhanced EdU incorporation in Ly6C^{lo} monocytes and CD16.2^{+} IM from RBP-J deficient mice (**Figure 6** – figure supplement 1A and B). These findings suggested that the augmented population of these cells might not be resulted from increased *in situ* proliferation. Previous reports have indicated that LPS can increase the population of IM in a CCR2-dependent manner(Sabatel et al., 2017). We thus challenged mice with LPS, and analyzed monocytes and IM at day 0 and day 4. LPS exposure induced an increase in numbers of Ly6C^{lo} monocytes, CD16.2^{+} IM and CD206^{-} IM compared to baseline both in control and RBP-J deficient mice (**Figure 6D and E**). There was a trend toward higher Ly6C^{hi} monocyte after LPS treatment, although this observation was not statistically significant. Of note, RBP-J deficient mice exhibited robustly elevated numbers of Ly6C^{lo} monocytes and CD16.2^{+} IM

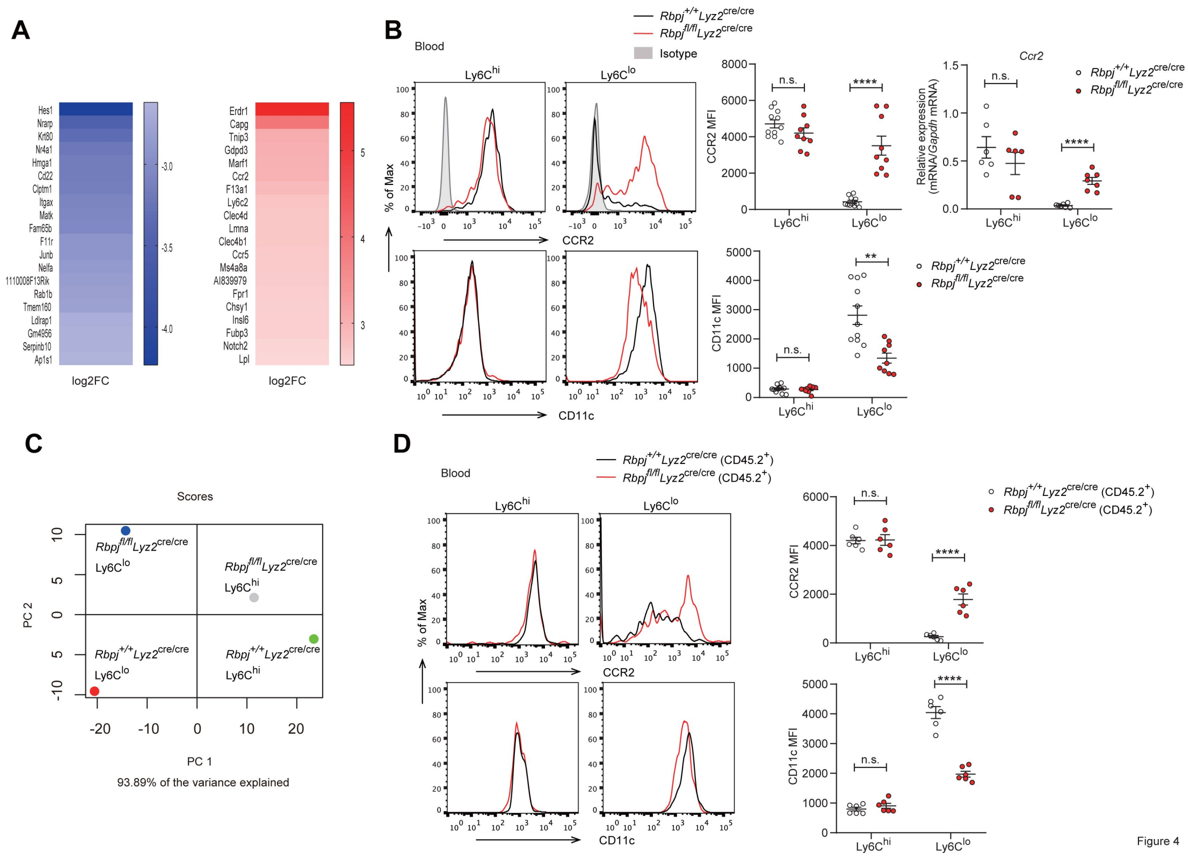

Figure 4.

Phenotypic markers in Ly6C^{lo} monocytes are changed in RBP-J deficient mice.

(A) Heatmap of RNA-seq dataset showing the top 20 downregulated and upregulated genes in blood Ly6C^{lo} monocytes from *Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$ versus *Rbpj*$^{+/+}$*Lyz2*$^{cre/cre}$ control mice. Blue and red font indicates downregulated and upregulated genes in RBP-J deficient Ly6C^{lo} monocytes respectively.
(B) Representative FACS plots, cumulative MFI, and qPCR analysis of CCR2/CD11c expression in control and RBP-J deficient blood monocyte subsets. Shaded curves represent isotype control, black lines represent control mice and red lines represent RBP-J deficient mice.
(C) PCA of indicated cell types.
(D) Representative FACS plots and cumulative MFI of CCR2/CD11c expression in blood monocyte subsets derived from control or RBP-J deficient mice. Black lines represent control mice, and red lines represent RBP-J deficient mice.
Data are pooled from 2 independent experiments (B and D); n ≥ 6 in each group. Data are shown as mean ± SEM; n.s., not significant; **, P < 0.01; ****, P < 0.0001 (two– tailed Student's unpaired t test). Each symbol represents an individual mouse.

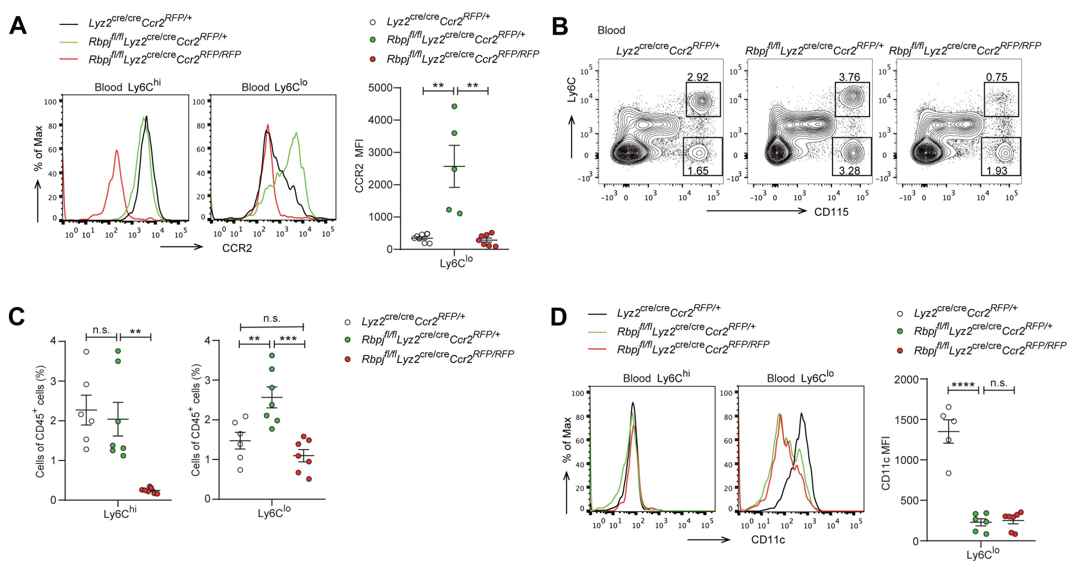

Figure 5.

Blood Ly6C^{lo} monocytes are decreased in DKO mice.

(A) Representative FACS plots and cumulative MFI of CCR2 expression in *Lyz2*cre/cre*Ccr2RFP/+* control, *Rbpjfl/flLyz2*cre/cre*Ccr2RFP/+* and *Rbpjfl/flLyz2*cre/cre*Ccr2RFP/RFP* (DKO) blood Ly6C^{hi} and Ly6C^{lo} monocytes are shown.
(B and C) Blood monocyte subsets in control, RBP-J deficient and DKO mice were determined by FACS. Representative FACS plots (B) and cumulative data of cell ratio
(C) are shown.
(D) Representative FACS plots and cumulative MFI of CD11c expression in control, RBP-J deficient and DKO blood Ly6C^{hi} and Ly6C^{lo} monocytes are shown.
Data are pooled from at least 2 independent experiments; n ≥ 5 in each group. Data are shown as mean ± SEM; n.s., not significant; **, $P < 0.01$; ***, $P < 0.001$; ****, $P < 0.0001$ (two-tailed Student's unpaired t test). Each symbol represents an individual mouse.

compared with control mice after LPS treatment (**Figure 6D and E**), suggesting that RBP-J deficient Ly6C^{lo} monocytes were recruited to inflamed tissue in large numbers and differentiated into CD16.2^{+} IM.

To investigate whether the elevated levels of lung monocytes and CD16.2^{+} IM in RBP-J deficient mice were derived from blood Ly6C^{lo} monocytes, we examined monocytes and IM in *Lyz2*$^{cre/cre}$*Ccr2*$^{RFP/+}$control, *Lyz2*$^{cre/cre}$*Ccr2*$^{RFP/RFP}$, *Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$*Ccr2*$^{RFP/+}$, and *Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$*Ccr2*$^{RFP/RFP}$(DKO) mice. The deletion of CCR2 led to a decrease in Ly6C^{hi} monocytes, and the Ly6C^{lo} monocytes in DKO mice were reduced to a level similar to that of control mice (**Figure 7A and B**). While the DKO mice had more CD16.2^{+} IM compared to control mice, these cells were severely reduced compared to RBP-J deficient mice (**Figure 7A and C**). The above data supported the notion that increased lung Ly6C^{lo} monocytes and CD16.2^{+} IM in RBP-J deficient mice were derived from Ly6C^{lo} blood monocytes. In summary, these results suggested that in RBP-J deficient mice, recruitment of blood Ly6C^{lo} monocytes to the lung was markedly facilitated by increased cell number as well as heightened expression of CCR2, leading to the increase of lung Ly6C^{lo} monocytes and CD16.2^{+} IM.

Discussion

In this study, we identified RBP-J as a pivotal factor in regulating blood Ly6C^{lo} monocyte cell fate. Our results showed that mice with conditional deletion of RBP-J in myeloid cells exhibited a robust increase in blood Ly6C^{lo} monocytes, and subsequently accumulated lung Ly6C^{lo} monocytes and CD16.2^{+} IM under steady-state conditions. Bone marrow transplantation experiments in which RBP-J deficient cells were transplanted into recipient mice as well as the parabiosis experiment showed similar increase in blood Ly6C^{lo} monocytes, demonstrating that RBP-J was an intrinsic factor required for the regulation of Ly6C^{lo} monocytes. Further analysis revealed that RBP-J regulated the expression of CCR2 and CD11c on the surface of Ly6C^{lo} monocytes. Moreover, the phenotype of elevated blood Ly6C^{lo} monocytes and their progeny was further ameliorated by deleting *Ccr2* in a RBP-J deficient background.

Notch-RBP-J signaling has been shown to play a role in regulating functional polarization and activation of macrophages, and regulated formation of Kupffer cells and macrophage differentiation from Ly6C^{hi} monocytes in ischemia(Foldi et al., 2016; Hu et al., 2008; Kang et al., 2020; Krishnasamy et al., 2017; Sakai et al., 2019; Wang et al., 2010; Xu et al., 2012). At steady-state, interaction of DLL1 with Notch 2 regulates conversion of Ly6C^{hi} monocytes into Ly6C^{lo} monocytes in special niches of the BM and spleen(Gamrekelashvili et al., 2016). Under inflammatory conditions, Notch2 and TLR7 pathways independently and synergistically promote conversion of Ly6C^{hi} monocytes into Ly6C^{lo} monocytes(Gamrekelashvili et al., 2020). However, in our study, the percentage of blood Ly6C^{lo} monocytes increased in RBP-J deficient mice. These results somewhat differ from what was observed in mice with conditional deletion of Notch2(Gamrekelashvili et al., 2016). Actually, 4 members of the Notch family have been identified in mammals(Radtke et al., 2013). Thus, Notch receptors seem to be non-redundant in regulating monocyte cell fate, and distinct Notch receptors are likely to act in a concerted manner to coordinate the monocyte differentiation program.

Colony-stimulating factor 1 receptor (CSF1R) signal, CX3CR1, CEBPβ and Nr4a1 have been suggested to be involved in the generation and survival of Ly6C^{lo} monocytes. The lifespan of Ly6C^{lo} monocytes is acutely shortened after blockade of CSF1R, and the numbers of these cells are reduced, and the percentage of dead cells increased in mice with endothelial cell-specific depletion of *Csf1*(Emoto et al., 2022; MacDonald et al., 2010). CX3CR1 and CEBPβ-knockout mice show accelerated death of Ly6C^{lo} monocyte, display the decreased level of Ly6C^{lo} monocytes(Landsman et al., 2009; Tamura et al., 2017). Nr4a1 knockout mice have significantly fewer monocytes in blood, bone marrow, and spleen, due to differentiation deficiency in MDP and accelerated

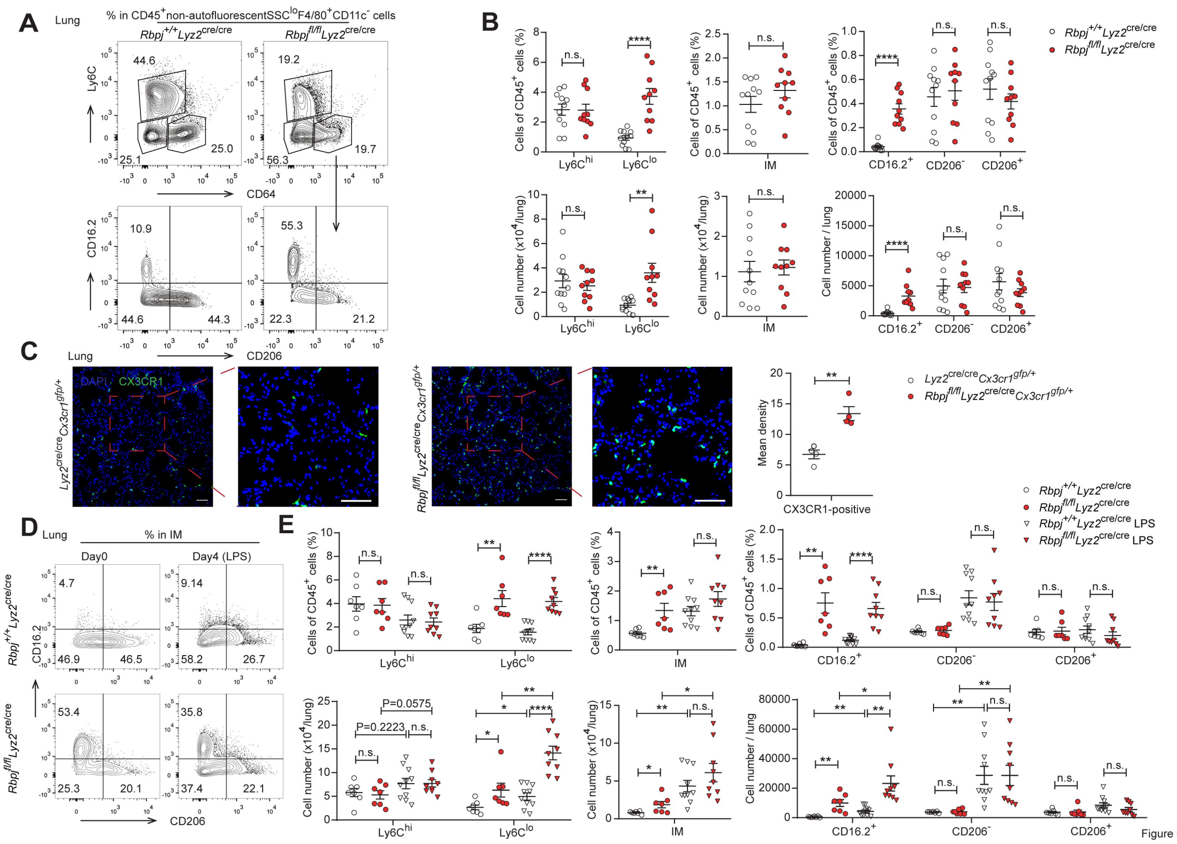

Figure 6.

RBP-J deficient mice exhibit more lung Ly6C^{lo} monocytes and CD16.2^{+} IM.

(A and B) Indicated populations in the lungs of *Rbpj$^{+/+}$Lyz2*$^{cre/cre}$ and *Rbpj$^{fl/fl}$Lyz2*$^{cre/cre}$ mice were determined by FACS. Representative FACS plots (A) and cumulative data of cell ratio and absolute numbers (B) are shown.

(C) Immunofluorescence staining for GFP^{+} cells in the lungs from *Lyz2*$^{cre/cre}$*Cx3cr1$^{gfp/+}$* and *Rbpj$^{fl/fl}$Lyz2*$^{cre/cre}$*Cx3cr1$^{gfp/+}$* mice (CX3CR1 [green]; DAPI [blue]). Scale bars represent 50 μm.

(D and E) *Rbpj$^{+/+}$Lyz2*$^{cre/cre}$ and *Rbpj$^{fl/fl}$Lyz2*$^{cre/cre}$ mice were instilled intranasally with PBS or PBS containing LPS, and lungs were harvested at the indicated time points. Representative FACS plots (D) and cumulative data of cell ratio and absolute numbers

(E) are shown.

Data are pooled from at least 2 independent experiments; n ≥ 4 in each group. Data are shown as mean ± SEM; n.s., not significant; *, P < 0.05; **, P < 0.01; ****, P < 0.0001 (two-tailed Student's unpaired t test). Each symbol represents an individual mouse.

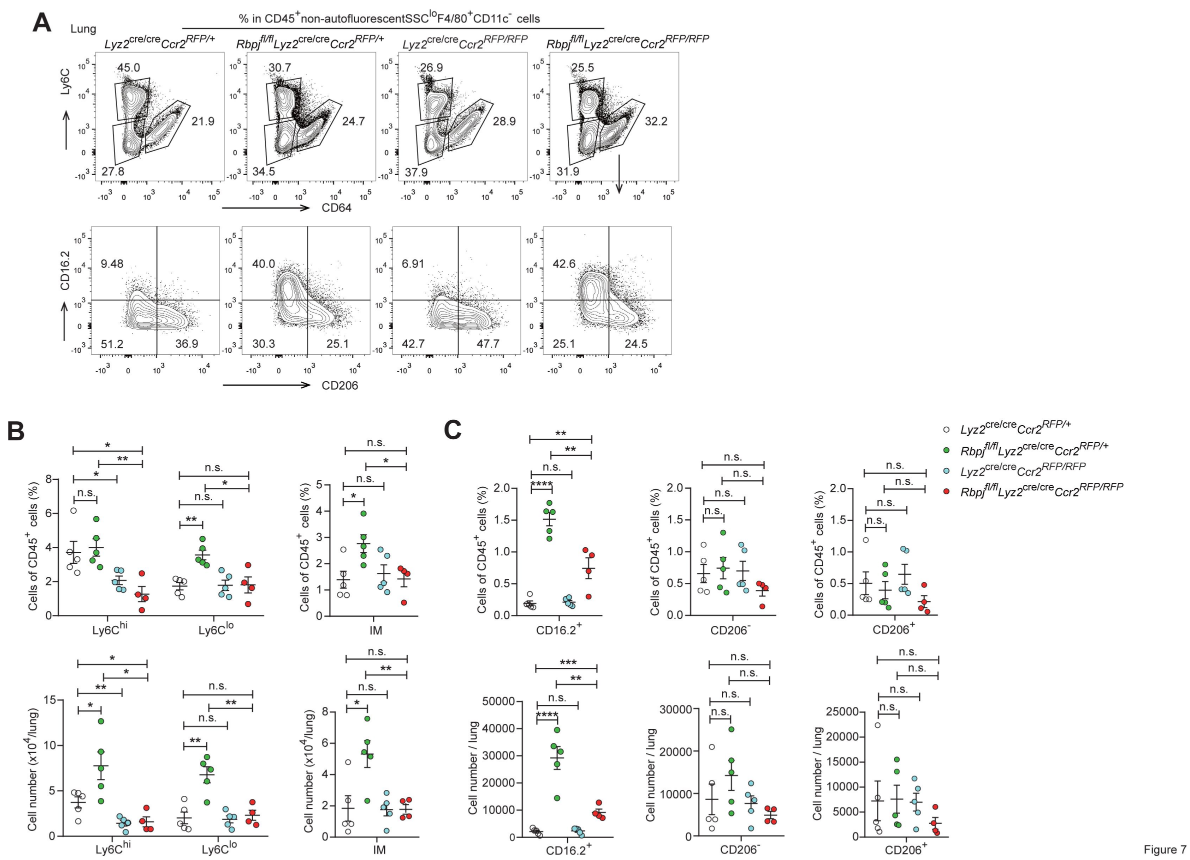

Figure 7.

DKO mice lack lung Ly6C^{lo} monocytes and CD16.2^{+} IM.

(A) Representative FACS plots of lung monocyte and IM subsets in *Lyz2*$^{cre/cre}$*Ccr2*$^{RFP/+}$, *Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$*Ccr2*$^{RFP/+}$, *Lyz2*$^{cre/cre}$*Ccr2*$^{RFP/RFP}$ and *Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$*Ccr2*$^{RFP/RFP}$ mice.
(B and C) Cumulative data of cell ratio and absolute numbers of monocyte (B) and IM
(C) subsets.
Data are pooled from 2 independent experiments; n ≥ 4 in each group. Data are shown as mean ± SEM; n.s., not significant; *, P < 0.05; **, P < 0.01; ***, P < 0.001; ****, P < 0.0001 (two-tailed Student's unpaired t test). Each symbol represents an individual mouse.

apoptosis of Ly6C^{lo} monocytes in bone marrow(Hanna et al., 2011). RBP-J deficient mice had normal MDP and Ly6C^{lo} monocytes, expressed normal level of Nr4a1 and Ki-67, exhibited normal half-life and turnover rate, which suggested that RBP-J may not regulate Ly6C^{lo} monocytes through these factors. Further studies are required to elucidate the precise molecular mechanisms of RBP-J in Ly6C^{lo} monocytes under steady state.

At the steady state, Ly6C^{lo} monocytes patrol endothelium of blood vessel. After R848 induced endothelial injury, Ly6C^{lo} monocytes recruit neutrophils to mediate focal endothelial necrosis and fuel vascular inflammation, and subsequently, the Ly6C^{lo} monocytes remove cellular debris(Carlin et al., 2013; Imhof et al., 2016). In response to *Listeria monocytogenes* infection, Ly6C^{lo} monocytes can extravasate but do not give rise to macrophages or DCs(Auffray et al., 2007). Additional evidence shows that Ly6C^{lo} monocytes do not merely act as luminal blood macrophages. In lung, exposure to unmethylated CpG DNA expands CD16.2^{+} IM, which originate from Ly6C^{lo} monocytes and spontaneously produce IL-10, thereby preventing allergic inflammation(Schyns et al., 2019). A recent study shows that Ly6C^{lo} monocytes give rise to CD9^{+} macrophages, which provide an intracellular replication niche for *Salmonella* Typhimurium in spleen, and Ly6C^{lo}-depleted mice are more resistant to *Salmonella* Typhimurium infection, suggesting that Ly6C^{lo} monocytes exert certain functions in systemic infection(Hoffman et al., 2021). Our data provide evidence for RBP-J in the function of blood Ly6C^{lo} monocytes, as RBP-J deficient Ly6C^{lo} monocytes exhibited enhanced competition in blood circulation, as well as gave rise to increased numbers of lung CD16.2^{+} IM. In summary, we showed that RBP-J acted as a fundamental regulator of the maintenance of blood Ly6C^{lo} monocytes and their descendent, at least in part through regulation of CCR2. These results provide insights into understanding the mechanisms that regulate monocyte homeostasis and function.

Acknowledgements

We thank the core facility at Institute for Immunology, Tsinghua University for valuable technical assistance.

Funding

This research was supported by National Natural Science Foundation of China grants (31821003, 31991174, 32030037 and 82150105 to X. Hu), a Ministry of Science and Technology of China grant (2020YFA0509100 to X. Hu), and funds from Tsinghua– Peking Center for Life Sciences and Institute for Immunology at Tsinghua University (to X. Hu).

Author contributions

T. Kou designed and performed experiments, analyzed and interpreted data, and wrote the manuscript. L. Kang designed and performed experiments, analyzed and interpreted data. B. Zhang provided advice on RNA-seq analyses. J. Li performed experiments. B. Zhao provided advice on experiments. W. Zeng provided advice on experiments. X. Hu conceptualized the project, supervised experiments, interpreted data, and wrote the manuscript.

Declaration of interests

The authors declare no competing interests.

Data availability

Sequencing data sets are deposited in the Genome Expression Omnibus with assigned accession numbers as follows: RNA-seq (GEO accession no. GSE208772).

Materials and methods

Mice

Cx3cr1$^{gfp/gfp}$ mice (JAX stock 005582; Jung et al., 2000) and *Ccr2*$^{RFP/RFP}$ mice (JAX stock 017586) were purchased from the Jackson Laboratory. Mice with a myeloid-specific deletion of the *Rbpj* were generated by crossing *Rbpj*$^{fl/fl}$ mice to *Lyz2*-Cre mice as described previously (Hu et al., 2008). *Rbpj*$^{fl/fl}$*Lyz2*$^{\text{cre/cre}}$ were crossed to *Cx3cr1*$^{gfp/gfp}$mice to obtain *Lyz2*$^{\text{cre/cre}}$*Cx3cr1*$^{gfp/+}$and *Rbpj*$^{fl/fl}$*Lyz2*$^{\text{cre/cre}}$*Cx3cr1*$^{gfp/+}$ mice. C*x3cr1*$^{gfp/+}$ mice were obtained by crossing *Cx3cr1*$^{gfp/gfp}$with C57/BL6 CD45.1^{+} mice. *Rbpj*$^{fl/fl}$*Lyz2*$^{\text{cre/cre}}$ mice were crossed with *Ccr2*$^{RFP/RFP}$ mice to obtain *Lyz2*$^{\text{cre/cre}}$*Ccr2*$^{RFP/+}$, $_{Lyz2}$cre/cre$_{Ccr2}$*RFP/RFP*, $_{Rbpj}$*fl/fl*$_{Lyz2}$cre/cre$_{Ccr2}$*RFP/+* and $_{Rbpj}$*fl/fl*$_{Lyz2}$cre/cre$_{Ccr2}$*RFP/RFP* mice. All mice were maintained under specific pathogen-free conditions. All animal experimental protocols were approved by the Institutional Animal Care and Use Committees of Tsinghua University. Gender– and age-matched mice were used at 7–12 weeks old for experiments.

Quantitative RT-PCR

Blood monocytes were sorted by flow cytometry. The total RNA was extracted using total RNA purification kit (GeneMarkbio) and reversely transcribed to cDNA by M– MLV Reverse Transcriptase (Takara). qPCR was performed on a real-time PCR system (StepOnePlus; Applied Biosystems) using FastSYBR mixture (CWBIO). *Gapdh* messenger RNA was used as internal control to normalize the expression of target genes. Primer sequences are provided in supplemental Table 1.

RNA-seq analysis

Ly6C$^{\text{hi}}$ and Ly6C$^{\text{lo}}$ blood monocytes were isolated from *Rbpj*$^{+/+}$*Lyz2*$^{\text{cre/cre}}$ and *Rbpj*$^{fl/fl}$*Lyz2*$^{\text{cre/cre}}$ mice, and total RNA was extracted using total RNA purification kit (GeneMarkbio). RNA was converted into RNA-seq libraries, which were sequenced with the pair-end option using an Illumina-HiSeq2500 platform at Beijing Genomics Institute (BGI), China. The significantly down-regulated genes were identified with P value <0.05 and (FPKM + 1) fold changes ≤0.2, and significantly up-regulated genes were identified with P value <0.05 and (FPKM + 1) fold changes ≥5.7. The RNA-seq data are deposited in Gene Expression Omnibus under accession number GSE208772

Annexin V staining

Annexin V and 7-amino-actinomycin D (7-AAD) were used for identification of apoptotic monocytes by flow cytometry. Blood cells were stained with annexin V (eBioscience) and 7-AAD (Biolegend) according to the manufacturers' protocols.

EdU pulsing and latex beads labeling

Mice were injected intravenously with a single 1 mg EdU (Thermo Scientific). BM and blood cells were collected and stained with Fluorescence-conjugated mAb against CD45, CD11b, Ly6G, CD115 and Ly6C. Cells were then fixed, permeabilized and stained with reaction cocktail using EdU Assay Kit. Labeled cells were analyzed by flow cytometry.

Gene	Forward primer	Reverse primer
Gapdh	ATCAAGAAGGTGGTGAAGCA	AGACAACCTGGTCCTCAGTGT
Rbpj	ACCCCTGTGCCTGTCGTAGA A	TCCCGGAATGCAGAAATGTC
Nr4a1	TTGAGCTTGAATACAGGGCA	AGTTGGGGGAGTGTGCTAGA
Ccr2	CCTTGGGAATGAGTAACTGT GTGAT	ATGGAGAGATACCTTCGGAAC TTCT

Table 1

Primers sequences for regular qPCR used in this study

Mice were injected intravenously with a single 10 µl latex beads (Polysciences) in 250 µl PBS. Blood cells were harvested at indicated time. Cells were then stained for CD45, CD11b, Ly6G, CD115, Ly6C, and analyzed by flow cytometry.

In vivo labeling of sinusoidal leukocytes

Mice were injected intravenously with 1 µg of PE-conjugated anti-CD45 antibodies. Two minutes after antibody injection, mice were sacrificed, and bone marrow were harvested. BM cells were then stained and analyzed by flow cytometry.

Generation of BM chimera

C57/BL6 CD45.2^{+} mice were lethally irradiated in two doses of 5.5Gy 2 h apart. 0.8 × 10^{5} BM cells from *Rbpj*$^{+/+}$*Lyz2*$^{cre/cre}$ mice (CD45.2^{+}) or *Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$ mice (CD45.2^{+}) were mixed with 3.2× 10^{5} BM cells from *Cx3cr1*$^{gfp/+}$ mice (CD45.1^{+}), and injected intravenously into recipient mice. Mice were used for experiments 8 weeks after irradiation.

Adoptive transfers

BM GFP^{+}Ly6C^{hi} monocytes were sorted from *Lyz2*$^{cre/cre}$*Cx3cr1*$^{gfp/+}$or *Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$*Cx3cr1*$^{gfp/+}$ mice, and transferred to *Ccr2*$^{RFP/RFP}$ mice. Blood cells were collected 60 hours later for flow cytometry analyses.

Parabiosis

Cx3cr1$^{gfp/+}$ CD45.1^{+} mice were surgically joined with age-matched female *Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$ or *Rbpj*$^{+/+}$*Lyz2*$^{cre/cre}$ CD45.2^{+} mice at the age of 6 weeks. Bloods were obtained via cardiac puncture, and cell populations were analyzed by flow cytometry at 4 weeks after the surgery.

Intranasal instillations of LPS

Rbpj$^{+/+}$*Lyz2*$^{cre/cre}$ or *Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$ mice were anesthetized with isoflurane and intranasally instilled with 10 µg LPS in 25 µl of PBS. Lungs were harvested 4 days later for flow cytometry analyses.

Immunofluorescence histology

Lungs from *Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$*Cx3cr1*$^{gfp/+}$ and *Lyz2*$^{cre/cre}$*Cx3cr1*$^{gfp/+}$ mice were fixed in 1% paraformaldehyde and incubated in 30% sucrose separately overnight at 4°C. The samples were then incubated in the mixture of 30% sucrose and OTC compound (Sakura Finetek) overnight at 4°C. The tissues were embedded and frozen in OCT compound and then cut at 10-µm thickness. Tissue sections were dried for 10 min at 50°C and then fixed in 1% paraformaldehyde at room temperature for 10 min, permeabilized in PBS/0.5% Triton X-100/0.3 M glycine for 10 min and blocked in PBS/5% goat serum for 1 h at room temperature. Sections were stained with rabbit anti-mouse GFP antibodies (1:1000; EASYBIO) overnight at 4°C, and washed with PBS/0.1% Tween-20 for 30 min three times at room temperature. Sections were then incubated with AF488-conjugated goat anti-rabbit antibodies (1:200; proteintech) for 2 h at room temperature, and washed with PBS/0.1% Tween-20 for 30 min three times at room temperature. Sections were stained with DAPI (Solarbio) for 7 min, washed in PBS for 8 min two times at room temperature and mounted with SlowFade Diamond Antifade Mountant (Life Technologies).

Cell isolation and flow cytometry

Peripheral blood (PB) was sampled by eyeball extirpating, spleens were mashed through a 70-µm strainer, and bone marrow cells were collected from femurs. Lungs were perfused with 5 ml of HBSS (MACGENE) through the right ventricle, excised, and digested in HBSS containing 5% FBS, 1 mg/ml collagenase type I (Sigma-Aldrich) and 0.05 mg/ml DNase I (Sigma-Aldrich) for 1 h at 37 °C.

The digested tissues were homogenized by shaking, passed through a 70-μm cell strainer to create a single cell suspension. The suspension was enriched in mononuclear cells and harvested from the 1.080:1.038 g ml^{-1} interface using a density gradient (Percoll from GE Healthcare). BM, Spleen, PB and lung cells were stained with fluorescently conjugated antibodies. The absolute number of cells was counted by using CountBright™ Absolute Counting Beads (Invitrogen).

Antibodies against CD45 (30-F11), Ly6C (HK1.4), CD64 (X54-5/7.1), CD206 (C068C2), CD16.2 (9E9), CD45.1 (A20), F4/80 (BM8), Biotin, CCR2 (SA203G11) and CD3ε (145-2C11) were purchased from BioLegend. Antibodies against Ly6G (1A8-Ly6g), CD11b (M1/70), CD115(AFS98), CD117(2B8), CD135 (A2F10), CD11c (N418), CD206 (MR6F3), CD45.2 (104), Nur77 (12.14), Ki-67 (SolA15) and lineage maker were purchased from eBioscience. Fluorescence-conjugated mAb against Ly6C (AL-21) were purchased from BD Biosciences. Isotype-matched antibodies (eBioscience) were used for control staining. All antibodies were used in 1:400 dilutions, and surface antigens were stained on ice for 30 min.

For intracellular staining, cells were stained with antibodies to surface antigens, fixed and permeabilized with the Cytofix/Cytoperm Fixation/Permeabilization Solution Kit (BD Biosciences), and stained with antibodies or isotype control diluted in permeabilization buffer separately for 30 min at room temperature.

Cells were analyzed on FACSFortessa or FACSAria III flow cytometer (BD Biosciences) using FlowJo software.

Statistical analysis

Statistical analysis was performed using GraphPad Prism software. All results are shown as mean ± SEM. Statistical significance was determined using Student's unpaired t test. Statistical significance was defined as $P < 0.05$.

Figure Legends

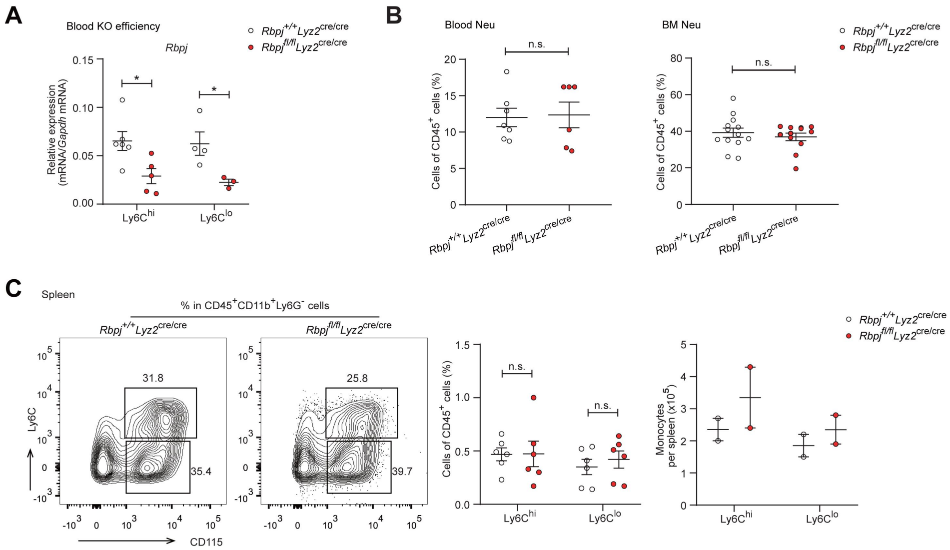

Figure 1 – Figure supplement 1.

Control and RBP-J deficient mice show similar neutrophils.

(A) qPCR analysis of *Rbpj* expression in sorted monocyte subsets from control and RBP-J deficient mice.
(B) Blood and BM CD45$^+$CD11b$^+$Ly6G$^+$ neutrophils were determined by FACS. Cumulative data of cell ratio are shown.
(C) Spleen monocyte subsets were determined by FACS. Representative FACS plots (left) and cumulative data of cell ratio and absolute numbers (right) are shown.
Data are pooled from at least 2 independent experiments; n ≥ 2 in each group. Data are shown as mean ± SEM; n.s., not significant; *, P < 0.05 (two-tailed Student’s unpaired t test). Each symbol represents an individual mouse.

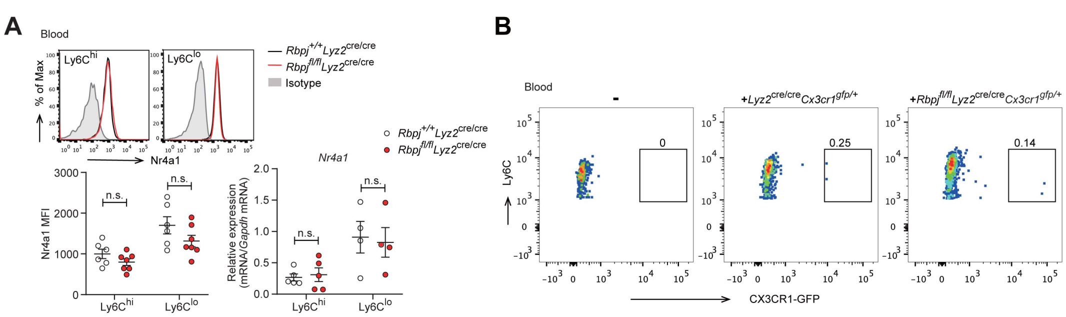

Figure 2 – Figure supplement 1.

The conversion of Ly6C^{hi} monocyte is identical in control and RBP-J deficient mice.

(A) Representative FACS plots, cumulative MFI and qPCR analysis of Nr4a1 expression in control and RBP-J deficient blood monocyte subsets. Shaded curves represent isotype control, black lines represent control mice, and red lines represent RBP-J deficient mice.
(B) Representative FACS plots of GFP$^+$Ly6C^{hi} monocytes in recipient.
Data are pooled from 2 independent experiments (A); n ≥ 4 in each group. Data are shown as mean ± SEM; n.s., not significant (two-tailed Student’s unpaired t test). Each symbol represents an individual mouse.

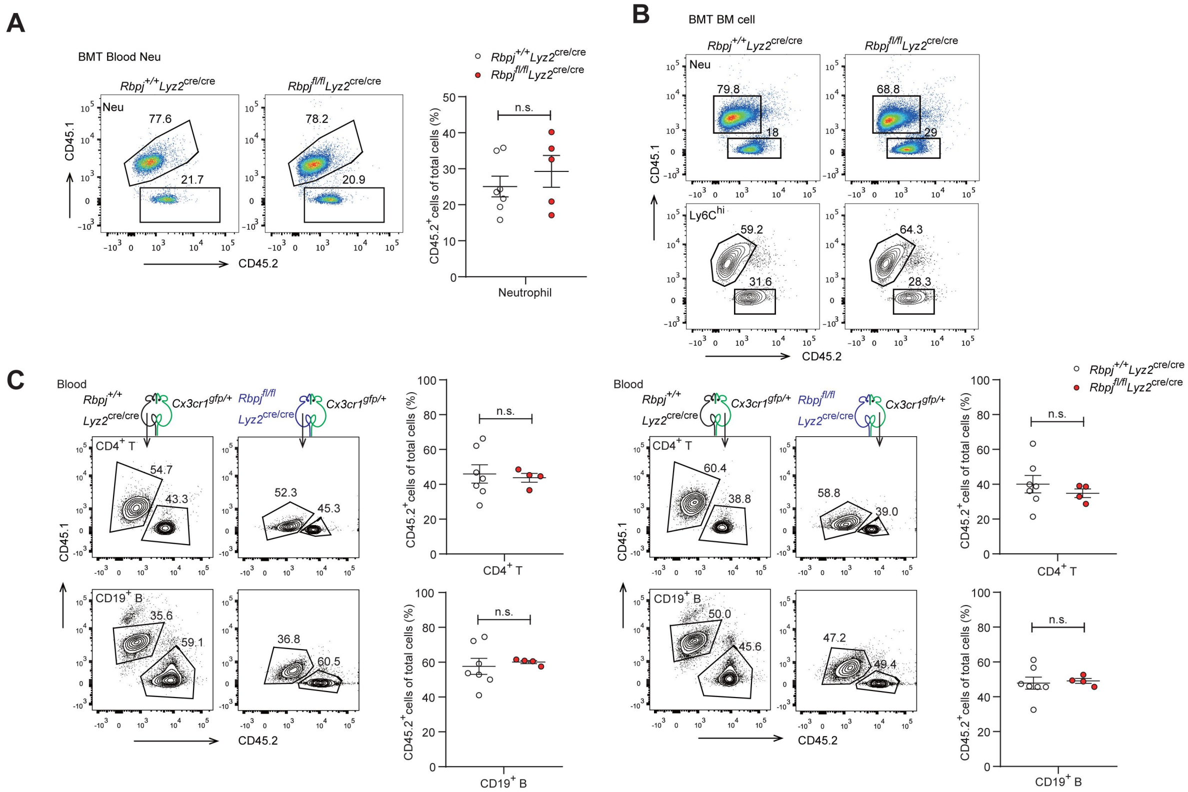

Figure 3 – Figure supplement 1.

Cell-intrinsic requirement of RBP-J for Ly6C^{lo} maintenance.

(A) Representative FACS plots (left) and cumulative data (right) quantitating percentages of blood neutrophils in recipient mice.

(B) BM neutrophils and Ly6C^{hi} monocytes in recipient mice were determined by FACS.

(C) CD4^{+} T cells and CD19^{+} B cells were analyzed by FACS in parabiotic mice.

Data are pooled from at least 2 independent experiments (A and C); n ≥ 4 in each group. Data are shown as mean ± SEM; n.s., not significant (two-tailed Student's unpaired t test). Each symbol represents an individual mouse.

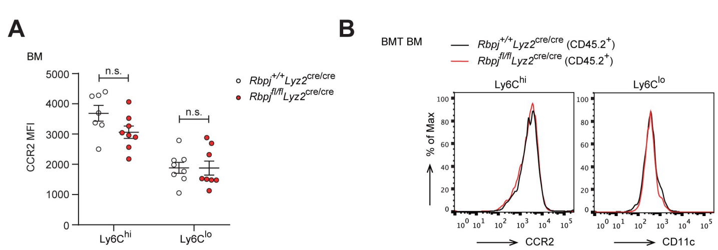

Figure 4 – Figure supplement 1.

Normal expression of CCR2 and CD11c in BM monocytes.

(A) Cumulative MFI of CCR2 expression in BM monocyte subsets.

(B) FACS plots of CCR2 and CD11c in BM Ly6C^{hi} and Ly6C^{lo} monocytes from control and RBP-J deficient mice.

Data are pooled from 3 independent experiments (A); n ≥ 7 in each group. Data are shown as mean ± SEM; n.s., not significant (two-tailed Student's unpaired t test). Each symbol represents an individual mouse.

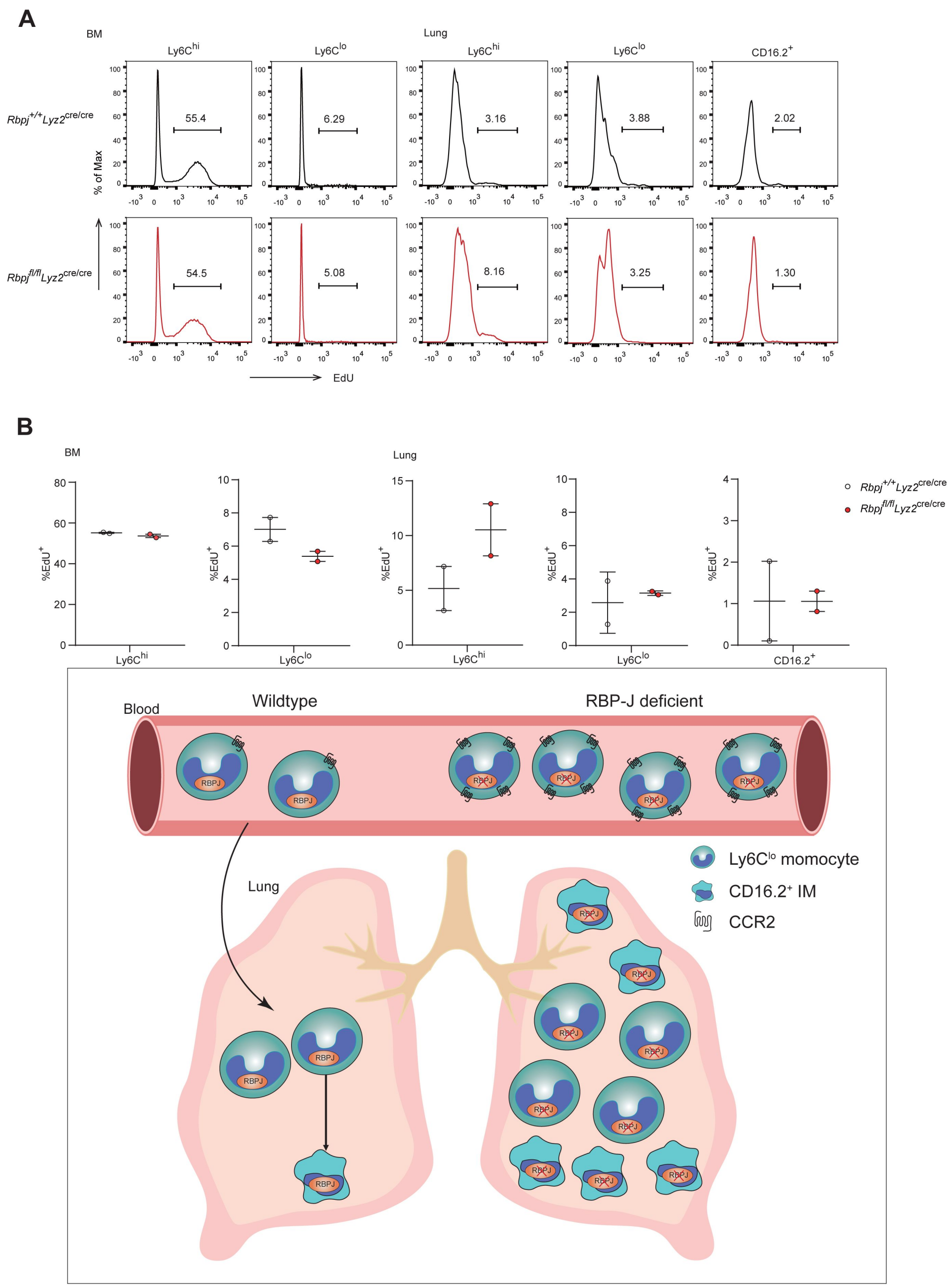

Figure 6 – Figure supplement 1.

RBP-J is not required for turnover of lung Ly6C^{lo} monocytes and CD16.2^{+} IM.

(A and B) Incorporation of EdU was assessed 24 hours after injection. BM monocyte subsets were used as controls. FACS plots (A) and representative data (B) are shown. Each symbol represents an individual mouse.

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Reviewer #1 (Public Review):

Kou and Kang et al. investigated the role of Notch-RBP-J signaling in regulating monocyte homeostasis. Specifically, they examined how a conditional knockout of Rbpj expression in monocytes though a Rbpjfl/fl Lyz2cre/cre mouse affects the homeostasis of Ly6Chi versus Ly6Clo monocytes. They discovered that Rbpj deficiency did not affect the percentage of Ly6Chi monocytes but instead, led to an accumulation of Ly6Clo monocytes in the peripheral blood. Using a comprehensive number of in vivo techniques to investigate the origin of this increase, the authors revealed that the accumulation of Rbpj deficient Ly6Clo monocytes was not due to an increase in bone marrow egress and homing and that this defect was cell intrinsic. However, EdU-labelling and apoptosis assays revealed that this defect was not due to an increase in proliferation nor conversion of Ly6Chi to Ly6Clo monocytes. Interestingly, it was revealed that Rbpj deficient Ly6Clo monocytes had increased expression of CCR2 and ablation of CCR2 expression reversed the accumulation of these cells in the periphery. Lastly, they discovered that Rbpj deficiency also led to downstream effects such as an accumulation of Ly6Clo monocytes in the lung tissue and increased CD16.2+ interstitial macrophages, presumably due to dysregulated monocyte differentiation and function.

Their findings are interesting and highlight a previously unexplored association between Notch-RBP-J signaling and CCR2 expression in monocyte homeostasis, providing further insight into the distinct pathways that regulate Ly6Chi vs Ly6Clo monocyte subsets individually.

The strengths of this paper include the use of multiple conditional genetic knock out mouse models to explore their hypothesis and the use of sophisticated in vivo techniques to study the major mechanisms involved in monocyte homeostasis. However, a major weakness of the paper is the exact role of how CCR2 compensates for the increase in Ly6Clo monocytes in the circulation in the RBP-J knockout mice as the authors showed no differences in their conversion, egress or homing back to the bone marrow. The authors were also unable to show that RBP-J knockout mice were functionally different in their response to CCL2 due to technical difficulties, which makes it challenging to conclude how CCR2 compensates for their trafficking patterns. Consequently the link between CCR2 and RBP-J remains correlative based on the data presented in the paper.

The conclusions of this paper are mostly well substantiated from the experimental data but as mentioned above, the mechanism of how CCR2 relates to the increase in Ly6Clo monocytes in RBP-J knockout mice is still unclear. Nevertheless, this work will be of interest to immunologists and biologists working on Notch-signalling in diseases. In addition, the methods and data would be useful for researchers who are seeking to use the Rbpjfl/fl Lyz2cre/cre mouse model for their studies.

https://doi.org/10.7554/eLife.88135.2.sa2

Reviewer #2 (Public Review):

The authors provide a compelling data to demonstrate that the Notch-related transcription factor RBP-J can influence the number of circulating and recruited monocytes. The authors first delete the Rbpj gene in the myeloid lineage (Lyz2) and show that, as a proportion, only Ly6Clo monocytes are increased in the blood. The authors then attempted to identify why these cells were increased in proportion but ruled out proliferation or reduced apoptosis. Next, they investigated the gene signature of Rbpj null monocytes using RNA-sequencing and identified elevated Ccr2 as a defining feature. Crossing the Rbpj null mice to Ccr2 null mice showed reduced numbers of Ly6Clo monocytes compared with Rbpj null alone. Finally, the authors identify that an increased burden of blood Ly6Clo monocytes is correlated with increased lung recruitment and expansion of lung interstitial macrophages.

The main conclusion of the authors, that there is a 'cell intrinsic requirement of RBP-J for controlling blood Ly6CloCCR2hi monocytes' is strongly supported by the data. However, other claims and aspects of the study require clarification and further analysis of the data generated.

Strengths
The paper is well written and structured logically. The major strength of this study is the multiple technically challenging methods used to reinforce the main finding (e.g. parabiosis, adoptive transfer). The finding reinforces the fact that we still know little about how immune cell subsets are maintained in situ, and this study opens the way for novel future work. Importantly, the authors have generated an RNA-sequencing dataset that will prove invaluable for identifying the mechanism - they have promised public access to this data via GEO - it is expected this will be made accessible upon publication.

Weaknesses - The main weakness of the study, is that although the main result is solidly supported, as written it is mostly descriptive in nature. For instance, there is no given mechanism by which RBP-J increases Ly6Clo monocytes. The authors conclude this is dependent on CCR2, however CCR2 deletion has a global effect on monocyte numbers and importantly in this study, it does not remove the Ly6Clo bias of cell proportions, if anything it seems to enhance the difference between the ly6C low and high populations in Rbpj null mice (figure 5C). This oversight in data interpretation likely occurred because: i) this experiment is missing a potentially important control (Lyz2cre/cre Ccr2RFP/RFP or RBP-J variations), and ii) lack of statistical comparisons between Ly6Clow and high subsets (e.g. two-way ANOVA design). In general, there seemed to be a focus on the Ly6C low cells, where the mechanism may be more identifiable in their precursors - likely the Ly6C high monocytes. Furthermore, the lack of this mechanism and data comparison may also be important, because it is possible that RBP-J signalling merely maintains the expression of Ly6C, rather than controls non-classical monocyte differentiation. In this case the comparison made for the sequencing data would be between Ly6C low non classical monocytes and 'artificially' Ly6C low classical monocytes. The basis of a population based on one marker is currently a widespread flaw in the field.

Other specific weaknesses were identified (note these are in addition to the more important comments above):

1. The confirmation of knockout in supplemental figure 1A shows only a two third knockdown when this should be almost totally gone. The authors have confirmed this is perhaps poor primer design and cite a study which shows almost complete reduction in protein levels (though this could be made more clear).

2. Many figures (e.g. 1A) only show proportional data (%) when the addition of cell numbers would also be informative - for example, what if Ly6Chigh cells were decreasing, thus artificially increasing the proportion of Ly6Clo cells? Looking at figure 7B - where cell numbers are shown, it is clear that cell proportion differences often do not match number data - here RBP-J knockout also increases Ly6C high cells in number (but not %).
3. It was noted previously that many figures only have an n of 1 or 2 (e.g. 2B, 2C), the authors clarified that some of these displayed one dot to represent an experiment of multiple n.
4. There is incomplete analysis (e.g. Network analysis, comparison to subset-restricted gene expression) and interpretation of RNA-sequencing results (figure 4), additionally the difference between the genotypes in both monocyte subsets would provide a more complete picture and potentially reveal mechanisms
5. The experiments in figure 5 are missing a control (Lyz2cre/cre Ccr2RFP/RFP or the Rbpj+/+ versions) and may have been misinterpreted. For example if the control (RBP-J WT, CCR2 KO) was used then it would almost certainly show falling Ly6C low numbers compared to RBP-J WT CCR2 WT, but RBP-J KO CCR2 KO would still have more Ly6c low monocytes than RBP-J WT, CCR2 KO - meaning that the RBP-J function is independent of CCR2. I.e. Ly6c low numbers are mostly dependent on CCR2 but this is irrespective of RBP-J. Explained in another way, the normal ratio of Ly6C high to low is around 1.5 Ly6Chigh cells for every one Ly6Clow cell, this is flipped in the RBP-J knockout to 1 high to 1.25 low (the main finding of the paper), but when CCR2 is removed it actually becomes 1 high to 5 low (actual numbers 0.2% vs around 1%) - in which case all CCR2 removal is doing is lowering the number of monocytes and RBP-J's mechanism is independent of CCR2.
6. Figure 6 was difficult to interpret because of the lack of shown gating strategy. The authors state they copied the strategy from Schyns et al. however in order to review this correctly the authors should show a supplemental figure of their own gating.
7. Figure 7 has the same problem as figure 5, but this time has the correct control. CCR2 ablation has a global suppression of monocyte numbers however the increased ly6c low monocyte ratio is most likely still present in the DKO mice - the lower numbers reduce the clarity of the data. Additionally in Lung IM macrophages depletion of CCR2 in the DKO only had a partial effect in some cell types - so CCR2 is playing a role, but it is not fully dependent. A good comparison would be if they blocked PU.1 expression - the effect of RBP-J would also be muted but it doesn't mean anything in terms of mechanism. Statements about the origin of the cells may need to be removed due to lack of compelling evidence.
8. Even after being notified and acknowledging the study, the authors still have not referred to or cited a similar 2020 study in their manuscript. This study also investigated myeloid deletion of Rbpj (Zhang et al. 2020 - https://doi.org/10.1096/fj.201903086RR). Zhang et al identified that Ly6Clo alveolar macrophages were decreased in this model - it is intriguing to synthesise these two studies and hypothesise that the ly6c low monocytes steal the lung niche, but this was not discussed. The authors also indicated they looked at AM but saw no difference - perhaps they should look specifically at Ly6Clow AMs in their data to compare with this study?

https://doi.org/10.7554/eLife.88135.2.sa1

Reviewer #3 (Public Review):

In this study, the authors investigate the role of the Notch signalling regulator RBP-J on Ly6Clow monocyte biology starting with the observation that RBP-J-deficient mice have increased circulating Ly6low monocytes. Using myeloid specific conditional mouse models,

the authors investigate how RBP-J deficiency effects circulating monocytes and lung interstitial macrophages.
A major strength of this study is that it provides compelling evidence that RBP-J is a novel, critical factor regulating Ly6Clow monocyte cell frequency in the blood. The authors demonstrate that RBP-J deficiency leads to increased Ly6Clow monocytes in the blood and lung and CD16.2+ interstitial macrophages in steady state. The authors use a number of different techniques to confirm this finding including bone marrow transplantation experiments and parabiosis.

The main conclusion of the paper is that RBP-J controls the fate of Ly6ClowCCR2hi monocytes in a cell-intrinsic manner. This conclusion is strongly supported by the data provided. However, this paper is predominantly descriptive and further research is required to fully uncover the mechanisms by which RBP-J deficiency leads to Ly6Clo monocyte numbers increasing specifically in the blood and lungs and the consequence of RBP-J deficiency on Ly6C-low monocyte functionality.

The authors have performed RNA-seq and more in-depth analysis of this sequencing may provide clues for uncovering the thus far elusive mechanism.

https://doi.org/10.7554/eLife.88135.2.sa0

Author Response

The following is the authors’ response to the original reviews.

We thank the reviewers for their time and insightful and constructive comments. We are pleased that reviewers found this study “opens the way for novel future work” and the findings “interesting”. We have experimentally addressed the points raised by the reviewers and have substantially revised the manuscript by modifying 30 figures panels. The reviewers’ points are specifically addressed below.

> 1. *The authors concluded that an accumulation of Ly6Clo monocytes occurred in the Rbpjfl/fl Lyz2cre/cre mouse by examining the percentage of cells among CD45+ cells in Figure 1. It would be helpful if the authors could give an account of the total cell count numbers of monocyte subsets per ml of blood and in the bone marrow to give the readers a better idea of the extent of increase as cell percentages among CD45+ cells may be influenced by the number of other immune subsets.*

We thank the reviewer for raising these points. In this research, we crossed Rbpjfl/fl mice with Lyz2-Cre mice carrying the Cre recombinase inserted in the Lysozyme-M (Lyz2) gene locus results in the selective deletion of RBP-J in myeloid cells, such as monocytes, macrophages and granulocytes. We then proceeded to examine the neutrophil levels in the bone marrow and blood. The percentage of neutrophils observed was found to be similar to that of control mice, which was in line with the findings reported in the literature (Metzemaekers et al. 2020). Furthermore, the proportion of Ly6Chi monocytes in RBP-J deficient mice was found to be similar to that of control mice, which is consistent with the literature (Ginhoux et al. 2014). Based on these results, we thought that the changes observed in the proportion of Ly6Clo monocytes could reliably indicate the alterations occurring in Ly6Clo monocytes within the Rbpjfl/flLyz2cre/cre mice.

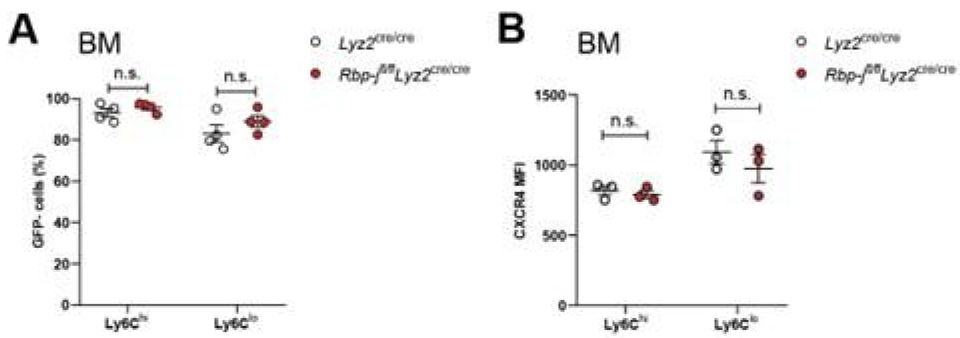
> 1. *The authors demonstrated no significant differences in bone marrow progenitor and monocyte numbers, therefore concluding that monocyte egress from the bone marrow did not contribute to the increase in Ly6Clo monocyte numbers in the blood (Figure 1B-D). As it is unclear what is the exact cell number increase in the blood, the changes in bone marrow monocyte numbers might be too small to be reflected in their percentage calculations. In light that CCR2 was also found to play a role in Ly6Clo monocyte homeostasis in Rbpjfl/fl Lyz2cre/cre mice, could the authors demonstrate if Rbpj-deficient Ly6Clo monocytes might be more responsive to CCL2 through transwell experiments? This would also provide readers a more in-depth mechanism of how an increase in CCR2 on Rbpj-deficient Ly6Clo monocytes leads to their accumulation in the periphery.*

The experimental results regarding the proportion of monocytes and precursor cells in the bone marrow were derived from multiple experiments. The data obtained from individual experiments as well as the final integrated data did not reveal significant differences between the control mice and Rbpjfl/flLyz2cre/cre mice. Therefore, we believed that even if there were small changes in cell numbers, these differences could still be reflected through alterations in their proportions. We attempted transwell experiments, but unfortunately, they were not technically successful. Nearly all sorted Ly6Clo monocytes attached to the transwell membrane, making it challenging to draw a conclusion regarding the responsiveness of RBP-J deficient Ly6Clo monocytes to CCL2.

> 1. *In the parabiosis experiment conducted in Figure 3C-E, the authors provide conclusive evidence that the accumulation of Rbpj-deficient Ly6Clo monocytes was cell intrinsic as Rbpj-deficient Ly6Clo monocytes continued to accumulate in the blood of control counterparts. Monocytes have also been shown to accumulate in the spleen and re-enter or home back to the bone marrow. Assessing if there is a change in monocyte homing abilities in Rbpj-deficient Ly6Clo monocytes by examining their numbers in the spleen and bone marrow of control parabiotic mice would substantiate their claims that the defect was cell intrinsic and provide further understanding for the readers of why Rbpj-deficient Ly6Clo monocytes accumulate in the blood.*

We thank the reviewer for bringing out this interesting point. We also analyzed the proportions of GFP- Ly6Chi monocytes and Ly6Clo monocytes in the bone marrow of parabiotic mice. The experimental results revealed that there were no significant differences in the proportion of GFP- monocytes between the control mice and the KO animals (see the figure A below). We also detected the expression of CXCR4 in bone marrow Ly6Clo monocytes. Rbpjfl/flLyz2cre/cre mice exhibited normal expression of CXCR4 (see Author response image 1 below), which participates in the homing of classical and nonclassical monocytes to bone marrow and spleen monocyte reservoirs (Chong et al. 2016). The homing abilities of RBP-J deficient Ly6Clo monocytes may not have changed.

Author response image 1.



> 1. *Authors should provide cell counts for Figure 5B to demonstrate the extent CCR2 depletion affects the number of Ly6Clo monocytes in Rbpjfl/fl Lyz2cre/cre mice as explained in point 1.*

As mentioned before, we believed that the proportion of circulating monocytes could, to some extent, provide evidence of the impact of CCR2 deficiency on Ly6Clo monocytes.

> ***Reviewer #2***
>
> 1. *The confirmation of knockout in supplemental figure 1A shows only a two third knockdown when this should be almost totally gone. Perhaps poor primer design, cell sorting error or low Cre penetrance is to blame, but this is below the standard one would expect from a knockout.*

Kang et al (PMID: 31944217) evaluated the knockout efficiency of Rbpj in sorted colonic macrophages of Rbp-jfl/flLyz2cre/cre mice using qPCR and immunoblotting. The qPCR result indicated a two-third knockdown, while the immunoblotting results demonstrated efficient deletion of RBP-J protein in Rbp-jfl/flLyz2cre/cre mice. As pointed out by the reviewer, the observed two-third knockdown, which is lower than the expected complete knockout, may be attributed to primer design.

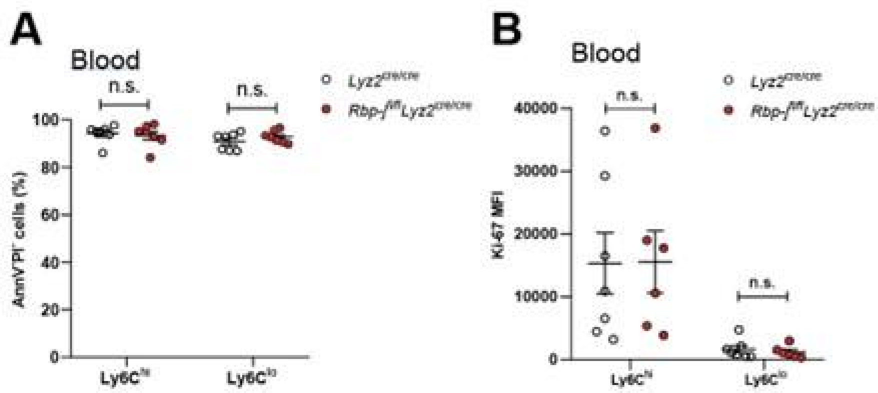
> 1. *Many figures (e.g. 1A) only show proportional data (%) when the addition of cell numbers would also be informative*

We appreciate the reviewer for bringing up these points. Indeed, multiple articles studying monocytes only show changes in cell proportions. As mentioned above, we believed that analyzing the proportion of circulating monocytes could offer valuable evidence of the influence of RBP-J deficiency on Ly6Clo monocytes.

> 1. *Many figures only have an n of 1 or 2 (e.g. 2B, 2C)*

Here, we employed annexin V (AnnV) and propidium iodide (PI) staining to evaluate apoptosis and cell death in Ly6Chi and Ly6Clo blood monocytes from control and RBPJ deficient mice. The results showed no significant difference in the levels of apoptosis and cell death between the two groups (see Author response image 2 below). The statistical data for Ki-67 expression obtained from multiple experiments, and the expression of Ki-67 showed no significant difference between the control and RBP-J deficient mice (see the figure B below). In Figure 2C, each dot represents 2-3 mice, and there were no differences observed between control and RBP-J deficient mice at multiple time points during the repeated measurements.

Author response image 2.



> 1. *Sometimes strong statements were based on the lack of statistical significance, when more n number could have changed the interpretation (e.g. 2G, 3E)*

We have derived the corresponding conclusions based on the observed experimental results.

> 1. *There is incomplete analysis (e.g. Network analysis) and interpretation of RNAsequencing results (figure 4), the difference between the genotypes in both monocyte subsets would provide a more complete picture and potentially reveal mechanisms*

We thank the reviewer for bringing out this point. We agreed that a more comprehensive analysis, including a comparison between the genotypes in both monocyte subsets, would provide a deeper understanding and potentially uncover underlying mechanisms. Having observed alterations in blood Ly6Clo monocytes in RBP-J deficient mice, our primary focus had been on analyzing the differentially expressed genes within this subset of monocytes to gain further insights into its specific characteristics and behavior. We also uploaded sequencing data sets in the Genome Expression Omnibus with assigned accession numbers GSE208772 to facilitate interested researchers in accessing and downloading the data.

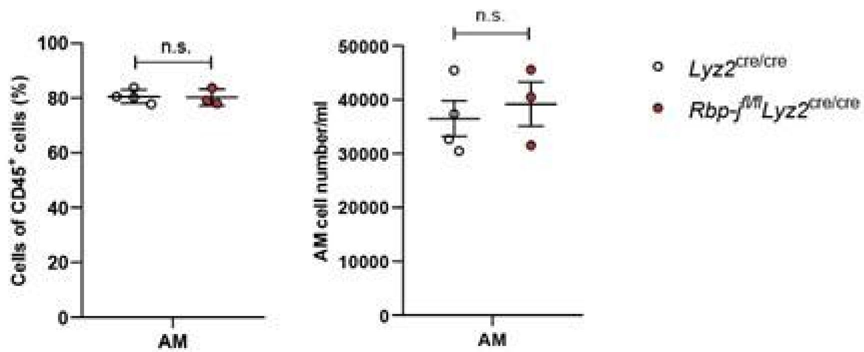
> 1. *The experiments in Figures 5 and 7 are missing a control (Lyz2cre/cre Ccr2RFP/RFP or the Rbpj+/+ versions) and may have been misinterpreted. For example if the control (RBP-J WT, CCR2 KO) was used then it would almost certainly show falling Ly6C low numbers compared to RBP-J WT CCR2 WT, but RBP-J KO CCR2 KO would still have more Ly6c low monocytes than RBP-J WT, CCR2 KO - meaning that the RBP-J function is independent of CCR2. I.e. Ly6c low numbers are mostly dependent on CCR2 but this is irrespective of RBP-J.*

The diminished Ly6Clo monocytes in Rbpjfl/flLyz2cre/creCcr2RFP/RFP (DKO) mice can be divided into two distinct subpopulations: one portion originates from Ly6Chi monocytes, while the other comprises Ly6Clo monocytes characterized by heightened CCR2 expression. The Ly6Clo monocytes that remain in DKO mice exhibit CCR2 expression levels within the normal range when compared to Lyz2cre/cre mice, but lower levels compared to RBP-J deficient mice (Figure 5A). These findings suggest that RBP-J exerts regulatory influence over Ly6Clo monocytes, at least in part, through CCR2.

> 1. *Figure 6 was difficult to interpret because of the lack of shown gating strategy. This reviewer assumes that alveolar macrophages were gated out of analysis*

The gating strategy of lung interstitial macrophage in the manuscript Figure 6 was consistent with the published work (Schyns et al, cited in the manuscript). We also measured alveolar macrophages (AM) from control and RBP-J deficient mice bronchoalveolar lavage fluid. At the resting state, RBP-J deficient mice exhibited normal AM frequency and number (see Author response image 3 below).

Author response image 3.



> 1. *The statements around Figure 7 are not completely supported by the evidence, i) a significant proportion of CD16.2+ cells were CCR2 independent and therefore potentially not all recently derived from monocytes, and ii) there is nothing to suggest that the source was not Ly6C high monocytes that differentiated - the manuscript in general seems to miss the point that the source of the Ly6C low cells is almost certainly the Ly6C high monocytes - which further emphasises the importance of both cells in the sequencing analysis*

Schyns et al and Sabatel at al showed that the numbers of IM and CD16.2+ were similar in Ccr2 sufficient and Ccr2-/- mice, demonstrating that CD16.2+ cells were Ccr2 independent. The number of CD16.2+ cells was significantly reduced in Rbpjfl/flLyz2cre/creCcr2RFP/RFP mice as compared to Rbpjfl/flLyz2cre/cre mice, in line with decreased number of lung Ly6Clo monocytes and blood Ly6Clo monocytes, showing that CD16.2+ cells depended on Ccr2 for their presence in Rbpjfl/flLyz2cre/cre mice.

> 1. *The authors did not refer to or cite a similar 2020 study that also investigated myeloid deletion of Rbpj (Qin et al. 2020 - https://doi.org/10.1096/fj.201903086RR). Qin et al identified that Ly6Clo alveolar macrophages were decreased in this model - it is intriguing to synthesise these two studies and hypothesise that the ly6c low monocytes steal the lung niche, but this was not discussed*

We thank the reviewer for bringing this study to our attention. According to their findings, myeloid-specific RBP-J deficiency resulted in a decrease in Ly6CloCD11bhi alveolar macrophages but an increase in Ly6CloCD11blo alveolar macrophages after bleomycin treatment, while the total number of alveolar macrophages showed no significant difference. These results suggest that RBP-J may play a role in regulating the balance between these specific alveolar macrophage subsets in response to bleomycin-induced injury, without affecting the overall population of alveolar macrophages. This may be different from what we observe in interstitial macrophages under resting conditions.

> ***Reviewer #3***
>
> 1. *It is curious that the authors do not see the increase in circulating monocytes reflected in the spleen however, the n-number is 2. Increasing the n-number would enable the author to understand the data which is not interpretable at the moment. There are multiple other places in which a low n-number makes it hard to fully understand the biology (eg Figure 2C&E)*

Although we only counted the number of splenic monocyte subsets in two mice, the proportion of splenic monocyte subsets was calculated based on additional quantity of mice in our study.

> 1. *Given that Ly6Clow monocytes are thought to be longer lived than Ly6C+ and there is still considerable labelling of Ly6Clow monocytes at the end of the 96 hours analysed in the EdU experiment, it is not possible to determine from the data here whether RBPJ deficiency increases life span. Could it be that differences in %EdU+ cells would only be seen at later time points? If the timeline was extended, could it be that differences in %EdU+ become apparent*

Based on the latex bead experiment, we observed that the presence of latex+ Ly6Clo monocytes at 7 days in control and RBP-J deficient mice did not differ, indicating that the lifespan of Ly6Clo monocytes did not increase.

> 1. *Similarly for the latex bead experiment. Given that there is only n=2 at the first time point and only ~30% of Ly6Clow monocytes are Latex+, it is very hard to conclusively claim that RBP-J does not influence monocyte survival or proliferation. An interesting experiment to assess whether RBP-J is increasing monocyte survival could be an adoptive transfer model in which Ly6Clow monocytes are injected into a congenic mouse and tracked over time.*

In RBP-J deficient mice, there was an increase in the proportion of Ly6Clo monocytes. We hypothesized that this lower proportion of latex+ cells might make it easier to observe differences, but clearly, in our experiment, no differences were observed between control and RBP-J deficient mice.

> 1. *RNA-seq: Ccr2 and Itgax are not the top hits. The authors do not investigate the top hits which may provide very interesting insight into how RBP-J influences monocyte biology.*

We thank the reviewer for raising these points. We also analyzed some top changed genes. The top two gene in the downregulated gene list are Hes1 and Nrarp, which are regulated by the Notch pathway (Krebs et al 2001 and Radtke et al 2010). We tested blood monocytes, but the population of monocyte subsets displayed no differences between Hes1fl/flRbp-jfl/flLyz2cre/cre and Rbp-jfl/flLyz2cre/cre mice (data not shown). As shown in Figure 2- figure supplement 1A, expression of Nr4a1 showed no significant differences between control and RBP-J deficient mice. The top gene in the upregulated gene list is Erdr1, which has been reported to play a role in cellular survival (Soto et al 2017), while blood monocyte subsets in RBP-J deficient mice displayed normal survival.

> 1. *The PCA plot in figure 4C- it would be interesting to see where all the biological replicates fall.*

We agree with the reviewer's assessment that observing the positions of all biological replicates on the PCA plot may indeed yield valuable insights. However, it is worth noting that the upregulated and downregulated genes also offer suggestive hints.

> 1. *Based on CCR2 expression and CD11c expression, monocytes from RBP-J deficient mice look more like Ly6C+ monocytes - could it be that RBP-J is increasing conversion from Ly6C+ monocytes to Ly6Clow? Or could it be that Ly6Clow monocytes are heterogeneous and RBP-J is increasing survival or conversion of one subtype of Ly6Clow monocytes but looking at all Ly6Clow monocytes together is masking this?*

Ly6Clo monocyte can be subdivided into different subpopulations depending on surface makers, such as CD43, MHC-II, CD11c and CCR2 (Jakubzick et al 2013 and Ginhoux et al. 2014). Carlin et al founded that a subset of blood Ly6Clow cells was independent of both Ccr2 and Nr4a1. As said by the reviewer, Ly6Clo monocytes are heterogeneous. Therefore, there is a possibility of altered survival in a certain group of Ly6Clo monocytes.

> 1. *The data presented here suggest that lung CD16.2+ interstitial macrophages are derived from Ly6Clow monocytes which are increased via CCR2. Although the data are suggestive, they are not conclusive, lineage tracing and CCR2 blockade or better, conditional CCR2 deficiency would help to strengthen the claim.*

Schyns et al showed that the number of CD16.2+ was similar in Ccr2 sufficient and Ccr2-/- mice, demonstrating that CD16.2+ cells were Ccr2 independent. While number of CD16.2+ cells was significantly reduced in Rbpjfl/flLyz2cre/creCcr2RFP/RFP mice as compared to Rbpjfl/flLyz2cre/cre mice, in line with decreased number of lung Ly6Clo monocytes and blood Ly6Clo monocytes. Moreover, the turnover of lung Ly6Chi and Ly6Clo monocytes was normal. These results implicated that CD16.2+ cells depended on Ccr2 for their presence in Rbpjfl/flLyz2cre/cre mice.

> 1. *The figures could do with more headings/ more detailed legends to help the reader, for example including what is BM, what is blood, what is spleen. Figure 2E needs the days labelled on or above the histograms.*

We thank the reviewer for raising this important point. We have now added additional detailed legends to the figure.

> 1. *Gating strategies should be included to help the reader understand which cells you are looking at, especially for Figure 6&7.*

The gating strategy for Figures 6 and 7 followed the method reported in the literature, which included the identification of alveolar macrophages. Additionally, we labeled the markers for cell populations in the figure.